

Space

Notes

Sputnik-1 :

- It was the first artificial Earth satellite.
- It was launched into space in an elliptical low Earth orbit (LEO) by Soviet Union on 4 October, 1957.
- It was an important incident in the history of space science.
- About 98 minutes were taken by this satellite to revolve one round of Earth.
- It was in orbit for three weeks before its batteries died, then silently revolved for two more months before fell back and burned into the Earth's atmosphere.
- It revolved around the Earth for 92 days and completed 1440 rounds of Earth.

Sputnik-2 :

- It was the second spacecraft of Soviet Union.
- It was launched into Earth orbit on 3 November, 1957.
- It was the first spacecraft to carry a living animal, a Soviet space dog named **Laika**, who died a few hours after the launch.

Vostok-I :

- It was the first spaceflight of the Vostok programme and the first manned spaceflight in history.
- It was launched from Baikonur Cosmodrome on April 12, 1961 with Soviet cosmonaut **Yuri Gagarin**, making him the first human to reach into outer space.
- He returned to Earth after successful completion of one round of Earth in 1 hour and 48 minutes.

First Landing on Moon :

- A Moon landing is the arrival of a spacecraft on the surface of Moon.
- This includes both unmanned & manned missions.
- The first human made object to reach the surface of the Moon was the Soviet Union's **Luna 2** mission, on 13 September, 1959.
- The man reached on Moon under Apollo program of U.S.A.
- Apollo Program crewed missions were started with Apollo-7 (1968) and ended with Apollo-17 (1972).
- **Apollo-11** was the spaceflight that landed the first two people on the Moon.
- Mission commander **Neil Armstrong** and pilot **Buzz Aldrin**, both American landed the lunar module Eagle on July 20, 1969 and walked on the lunar surface.

- **Michael Collins** piloted the command module Columbia alone in lunar orbit.
- Armstrong and Aldrin spent 21.5 hours on the lunar surface.

Launching Vehicles of India :

- The Satellite Launch Vehicle or SLV was a project started in the early 1970s by the Indian Space Research Organisation (ISRO) to develop the technology needed to launch satellites. The project was headed by APJ Abdul Kalam. SLV was intended to reach a height of 400 kilometres and carry a payload of 40 kilograms.

(i) Satellite Launch Vehicle-3 (SLV-3) :

- It was the India's first experimental satellite launch vehicle.
- It was an all solid, four stage vehicle weighing 17 tonnes with a height of 22m.
- It was capable of placing about 40 kg. class payloads in LEO.
- It was successfully launched on July 18, 1980 from Sriharikota Range (SHAR), when Rohini Satellite, RS-1 was placed in orbit.
- India became the sixth member of an exclusive club of space-faring nation with this achievement.
- RS-D1 was launched by SLV-3 from SHAR center on 31 May, 1981 and RS-D2 was launched on 17 April, 1983.

(ii) Augmented Satellite Launch Vehicle (ASLV) :

- It was a five stage, all solid-fuel rocket. It was developed by the ISRO.
- It was capable to place 150 kg satellite into LEO (i.e. 400 km).
- Its height was 24 metres and lift off weight was 40 tonnes.
- Its first developmental flight took place on 24 March, 1987 and the second on 13th July, 1988. The third developmental flight, ASLV-D3 was launched on 20 May, 1992, when SROSS-C (106 kg) was put into orbit. ASLV-D4 launched on 4 May, 1994, orbited SROSS-C2 (106 kg).
- After, that the SLV-III and ASLV Rocket launching series have been declared closed.

(iii) Polar Satellite Launch Vehicle (PSLV) :

- The Polar Satellite Launch Vehicle is an expendable (used only once) medium-lift launch vehicle.
- It was designed and operated by the ISRO.
- It is 44 metre long with a mass 295 tonnes.
- It is four stage third generation launch vehicle of India. It is the first Indian launch vehicle to be equipped with liquid stages.

- In first and third stage it uses solid propellants while in 2nd & 4th stage liquid propellants are used.
 - Propellants have two parts-
(a) Fuel - in which combustion takes place.
(b) Oxidiser - It assists in combustion of fuel (Provides oxygen for combustion).
 - Six boosters are attached with first stage of rocket. These are filled with solid propellants. The propellants amount is adjusted according to the requirements.
 - Hydroxyl Terminated Polybutadine (HTBP) is used as solid propellant in core motor of first stage and attached boosters with it.
 - The second stage of PSLV is operated by liquid propellant in which unsymmetrical Dimethyl Hydrazine-UDMH is used as fuel and N_2O_4 (Nitrogen Tetra Oxide) is used as oxidiser.
 - The third stage is operated by solid propellant. HTBP is used in this stage.
 - In the 4th & last stage as liquid propellant - Monomethyl Hydrozine (MMH) and as an oxydiser - mixed oxides of Nitrogen (MON-3) is used.
 - The first mission of PSLV-D1 (20 September, 1993) was unsuccessful. After its first successful launch on 15 Oct. 1994 (PSLV-D2), PSLV emerged as the reliable and versatile '**workhorse**' launch vehicle of India with 62 missions (in which 60 are successful) till December, 2024.
 - Most notable among these was the launch of PSLV-C37 on 15 February 2017, successfully deploying 104 satellites in sun-synchronous orbit, which was a record for the highest number of satellites sent to space on a single launch, until 24 January 2021, when SpaceX launched the Transporter-1 mission on a Falcon 9 rocket carrying 143 satellites into orbit.
 - This vehicle successfully launched two spacecrafts (both with XL version) – Chandrayaan-1 in 2008 and Mars Orbiter Spacecraft (Mangalyaan) in 2013 – that later traveled to Moon and Mars respectively. PSLV also launched India's first space observatory, Astrosat and India's first solar mission, Aditya-L1.
 - Core Alone is PSLV lightest version without six strap-on motors (boosters). It is used for launching smaller payloads i.e. 500-700 kg payload satellites.
 - The boosters attached with first stage of rocket are used when satellites with approximate payload of 1500 kg are launched. This version is known as **standard version**. It can take upto 1,750 kg of payload to Sun-Synchronous Polar Orbits of 600 km altitude.
 - In its boosters nine tonne solid propellant is used.
 - **PSLV-XL** (PSLV-extra large; Lift off mass - 320 tonnes) is an extended version of PSLV. In its boosters twelve tonne propellant is used. It was first used to launch Chandrayaan-I in its orbit.
 - **PSLV-DL** is another variant of PSLV (with two strap-on motors), which was first launched on 24 January, 2019.
 - Latest variant of this vehicle is **PSLV-QL** (with four Strap-on motors) which was first launched on 1 April, 2019.
 - Due to its unmatched reliability, PSLV has also been used (generally in XL version) to launch various satellites into Geosynchronous and Geostationary Orbits, like satellites from the IRNSS constellation.
 - At present, PSLV is configured with four variants like 6, 4, 2 solid rocket strap-on motors and core alone versions. Variants will be chosen based on the payload weights and orbit to be accomplished.
- (iv) **Geosynchronous Satellite Launch Vehicle (GSLV) :**
- Geosynchronous Satellite Launch Vehicle (GSLV) is an expendable launch system operated by ISRO.
 - PSLV is unable to launch satellites of payload more than 2500-3000 kg in Geostationary Transfer Orbits (GTO).
 - So, India has developed more powerful rockets (i.e. GSLV) to launch massive payloads satellites.
 - GSLV has three models :
(a) GSLV : Payload capacity is 1800 kg (to GTO).
(b) GSLV-MkII : Payload capacity is 2250 kg (to GTO).
(c) GSLV-MkIII : Payload capacity is 4000 kg (to GTO).
 - GSLV-MkII is a three stage vehicle with four liquid strap-ons. Its height is 49.13m and weight is 414.75 tonnes. The indigenously developed Cryogenic Upper Stage (CUS) forms third stage of GSLV-MkII.
 - GSLV-MkIII (now renamed as LVM3) is a three stage heavy lift launch vehicle which has two solid strap-ons, a core liquid booster and a CUS. Its height and weight is 43.43 m and 640 tonnes respectively. It is designed to carry 4 ton class of satellites into GTO or about 10 tons of payload to LEO, which is about twice the capability of GSLV Mk II.
 - Chandrayaan-2 was launched on its mission to the Moon from Satish Dhawan Space Centre SHAR, Sriharikota on 22 July, 2019 by GSLV-Mk III (LVM3-M1).
 - The first commercial mission of GSLV-Mark III (LVM3-M2) was successfully launched on 23 October, 2022 from Satish Dhawan Space Centre, Sriharikota. This was the fifth flight of LVM3. A total of 36 OneWeb Gen-1 satellites of about 150 kg each totalling about 5796 kg were launched to a circular low-Earth orbit (LEO) of

about 601 km in this mission. It was the first launch of LVM3 to LEO and the first Indian rocket with six tonne payload.

- Chandrayaan-3 was launched on its mission to the Moon from SDSC, SHAR, Sriharikota on 14 July, 2023 by LVM3-M4.
- GSLV-MkIII (now renamed as LVM3) is also known as **fatboy** due to its weight equal to fully filled 5 Boeing Jet aeroplanes or 200 elephants.
- Previously scientists have named PSLV as **workhorse** and GSLV-II as **naughty boy**.

(v) Small Satellite Launch Vehicle (SSLV)

- SSLV is being developed with complete indigenous technologies to meet the small satellite launch market on demand driven basis.
- It is a 3-stage launch vehicle configured with three solid propulsion stages and liquid propulsion based Velocity Trimming Module (VTM) as a terminal stage.
- SSLV is 2m in diameter and 34 m in length with lift off weight of ~ 120 tonnes. It is capable of launching ~ 500 kg satellite in 500 km low-Earth orbit (LEO) or ~ 300 kg satellite in 500 km Sun-synchronous orbit.
- The key features of SSLV are low cost, with low turn-around time, flexibility in accommodating multiple satellites, launch on demand feasibility, minimal launch infrastructure requirements etc.
- The first developmental flight of this launch vehicle (SSLV-D1), carrying a Earth Observation Satellite (EOS-02) and a student-made satellite AzaadiSAT was launched on 7 August, 2022 from Satish Dhawan Space Centre SHAR, Sriharikota. However, SSLV-D1 placed these satellites into an elliptical orbit instead of a circular one, so they were no longer usable.
- The second developmental flight of this launch vehicle (SSLV-D2) was launched successfully on 10 February, 2023 from SDSC SHAR, Sriharikota. It injected EOS-07 (a 156.3 kg Earth Observation Satellite of ISRO), Janus-1 (a 10.2 kg satellite belongs to ANTARIS, USA) and AzaadiSAT-2 (a 8.7 kg satellite which is a combined effort of about 750 girl students across India guided by Space Kidz India, Chennai) into 450 km circular LEO.
- The third developmental flight of this launch vehicle (SSLV-D3) was launched successfully on 16 August, 2024, from SDSC SHAR, Sriharikota. The SSLV-D3 placed EOS-8 (a 175.5 kg Earth Observation Satellite of ISRO)

precisely into the 475 km circular LEO. The mission also carried SR-0 DEMOSAT passenger satellite. This flight completed the SSLV Development Project and enables operational missions by Indian industry and **NewSpace India Limited (NSIL)**.

- ISRO is developing a new dedicated launch site in **Kulasekharapattinam**, a coastal village in Thoothukudi district of Tamil Nadu, which will handle SSLV launches. PM Narendra Modi remotely laid the foundation stone for India's this second spaceport on 28 February, 2024.

Vikram-S :

- Vikram-S, India's first privately built rocket was launched on 18 November, 2022 by Indian Space Research Organisation from Sriharikota. The mission for the rocket launch has been named as 'Prarambh', meaning beginning.
- As a tribute the rocket has been named after Vikram Sarabhai, who is considered as father of India's space programme.
- Vikram-S rocket was developed by Skyroot Aerospace and was assisted for launch by ISRO and IN-SPACe (Indian National Space Promotion and Authorisation Centre).

Cryogenics :

- In physics, cryogenics is the production and behaviour of materials at very low temperature.
- It is not well defined at what point on the temperature scale refrigeration ends and cryogenics begins, but scientists assume a gas to be cryogenic if it can be liquified at or below -150°C (123 K ; -238°F).
- A person who studies elements that have been subjected to extremely cold temperatures is called cryogenicist.
- This word has been derived from Greekword-kryos—and genic—, which means to produce thing as cold as ice.
- This word was firstly used by Kammerlingh Onnes in 1894, relating to liquifaction of Oxygen, Nitrogen, Hydrogen & Helium gas.
- Some coldest cryogenic liquids are as follows :

S.No.	Fluid	Boiling temperature Celcius	Boiling temperature Fahrenheit
1.	Oxygen	-183°	-297°
2.	Nitrogen	-196°	-320°
3.	Neon	-246°	-411°
4.	Hydrogen	-253°	-423°
5.	Helium	-270°	-452°

- Cryogenic technique is mainly used in rockets where gases are used as fuel in their liquid form.
- Generally Hydrogen and Oxygen are used in liquid form as fuel and oxidiser respectively in cryogenic engine.

Ramjet :

- Ramjet sometimes referred to as a flying stovepipe or an athodyd (an abbreviation of aero thermodynamic duct).
- It is a form of air breathing jet engine, that uses the engine's forward motion to compress incoming air without an axial compressor or a centrifugal compressor.
- Ramjets can be operated on supersonic speeds with Mach 5.

Scramjet :

- It is also known as Supersonic Combustion Ramjet in which combustion takes place in supersonic airflow.
- As in Ramjet, a Scramjet relies on high vehicle speed to compress the incoming air, but whereas a Ramjet decelerates the air to subsonic velocities before combustion, the airflow in a Scramjet is supersonic throughout the entire engine. That allows the Scramjet to operate efficiently at extremely high speeds.
- In Scramjet engine developed by ISRO, Hydrogen is used as fuel and Oxygen is used from the atmosphere as the oxidizer.
- The first experimental mission of ISRO's Scramjet engine towards the realisation of an Air Breathing Propulsion System was successfully conducted on 28 August, 2016 from Satish Dhawan Space Centre SHAR, Sriharikota. This test was the maiden short duration experimental test of ISRO's Scramjet engine with a hypersonic flight at Mach 6. ISRO's Advanced Technology Vehicle (ATV), which is an advanced sounding rocket, was the solid rocket booster used for the test of Scramjet engine at supersonic conditions.

Reusable Launch Vehicle - Technology Demonstrator (RLV-T.D.):

- It is one of the most technologically challenging endeavors of ISRO towards developing essential technologies for a fully reusable launch vehicle to enable low cost access to space.
- The configuration of RLV-TD is similar to that of an aircraft and combines the complexity of both launch vehicles and aircraft. The winged RLV-TD has been configured to act as a flying test bed to evaluate various technologies, namely, hypersonic flight, autonomous landing and powered cruise flight. In future, this vehicle will be scaled up to become

the first stage of India's reusable two stage orbital launch vehicle.

- RLV-TD was successfully flight tested on 23 May, 2016 from SDSC SHAR, Sriharikota validating the critical technologies such as autonomous navigation, guidance & control, reusable thermal protection system and re-entry mission management.

Sounding Rocket :

- A sounding Rocket, sometimes called a research rocket, is an instrument carrying rocket designed to take measurements and perform scientific experiments during its sub-orbital flight.
- It is capable to carry research apparatuses for measurement and research work upto a height of 50 to 1500 km above Earth level.

Hubble Space Telescope :

- The Hubble Space Telescope (often referred to as HST or Hubble) is a space telescope that was launched into low Earth orbit in 1990 and remains in operation.
- The Hubble telescope is named after astronomer Edwin Hubble and is one of NASA's Great Observatories, along with the **Compton Gamma Ray Observatory** (1991-2000), the **Chandra X-ray Observatory** (1999-present), and the **Spitzer Space Telescope** (2003-2020).
- The telescope completed 35 years of operation in April, 2025 and is predicted to last until 2030 to 2040.

Compton Gamma Ray Observatory (CGRO) :

- It was a space observatory detecting photons with energies from 20 keV to 30 GeV, in Earth orbit from 1991 to 2000.
- The observatory was launched from Space Shuttle Atlantis during STS-37 on April 5, 1991, and operated until its deorbit on June 4, 2000.

Chandra X-ray Observatory :

- The Chandra X-ray Observatory (CXO), previously known as the Advanced X-ray Astrophysics Facility (AXAF), is a Flagship-class space telescope launched aboard the Space Shuttle Columbia during STS-93 by NASA on July 23, 1999.
- Chandra is an Earth Satellite in a 64-hour orbit, and its mission is ongoing as of 2024.
- The telescope is named after the Nobel Prize-winning Indian-American astrophysicist Subrahmanyan Chandrasekhar.

Spitzer Space Telescope :

- Formerly it was known as Space Infrared Telescope Facility.

- It was launched on 25 August, 2003 by Delta Rocket from Cape Canaveral and retired on 30 January, 2020.
- It was the fourth and final of the NASA's **Great Observatories** programme.
- Each observatory of this programme has been launched into space for surveying the different radiations as visible, gamma, X-ray & infrared.
- Their main purpose is to study the early universe, young galaxies and forming stars and is used to detect dust disks around stars, considered an important signpost of planetary formation.

Kepler Space Telescope :

- The Kepler space telescope was launched by NASA on Delta II rocket in 2009 to discover Earth-sized planets orbiting other stars (exoplanets).
- Named after astronomer Johannes Kepler, the spacecraft was launched into an Earth-trailing heliocentric orbit.
- After nine and a half years of operation, the telescope's reaction control system fuel was depleted, and NASA announced its retirement on October 30, 2018. Kepler observed more than 530,500 stars and detected more than 2,600 exoplanets.

TESS :

- Transiting Exoplanet Survey Satellite (TESS) is a space telescope for NASA's Explorer program, designed to search for exoplanets using the transit method in an area 400 times larger than that covered by the Kepler mission.
- It was launched on 18 April 2018, atop a Falcon 9 launch vehicle.

ASTROSAT :

- ASTROSAT is India's first dedicated multi-wavelength space observatory. It was launched on a PSLV-XL on 28 September, 2015.
- This scientific satellite mission endeavours for a more detailed understanding of our universe. One of the unique features of ASTROSAT mission is that enables the simultaneous multi-wavelength observations of various astronomical objects with a single satellite.
- ASTROSAT observes universe in the optical, Ultraviolet, low and high energy X-ray regions of the electromagnetic spectrum, whereas most other scientific satellites are capable of observing a narrow range of wavelength band.
- Multi-wavelength observations of ASTROSAT can be further extended with co-ordinated observations using other spacecraft and ground based observations. All major astronomy institutions and some universities in India are participating in these observations.

XPoSat :

- XPoSat (X-ray Polarimeter Satellite) is the first dedicated scientific satellite from ISRO to carry out research in space-based polarisation measurements of X-ray emission from celestial sources. ISRO's PSLV-C58 has launched XPoS at Satellite on 1 January, 2024.
- The spacecraft carried two scientific payloads in a low-Earth orbit. The primary payload POLIX (Polarimeter Instrument in X-rays) will measure the polarimetry parameters (degree and angle of polarization) in medium X-ray energy range of 8-30 keV photons of astronomical origin. The XSPECT (X-ray Spectroscopy and Timing) payload will give spectroscopic information in the energy range of 0.8-15 keV.
- POLIX is realized by Raman Research Institute and XSPECT is by Space Astronomy Group of URSC.

James Webb Space Telescope :

- The James Webb Space Telescope (JWST) is developed by NASA in collaboration with the European Space Agency (ESA) and the Canadian Space Agency (CSA). It is designed primarily to conduct infrared astronomy.
- The JWST was launched on 25 December, 2021 from Kourou, French Guiana with ESA's Ariane 5 rocket.
- It is intended to succeed the Hubble Space Telescope as NASA's flagship mission in astrophysics.
- It is named after James E. Webb, who was the administrator of NASA from 1961 to 1968.
- In July, 2022, JWST's commissioning process was complete and it began transmitting its first data.
- The first JWST image (Webb's First Deep Field) was released to the public on 11 July, 2022.

International Space Station :

- The International Space Station (ISS) is a large space station assembled and maintained in low-Earth orbit by a collaboration of five space agencies : NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), CSA (Canada), and their contractors. ISS is the largest space station ever built.
- Its primary purpose is performing microgravity and space environment experiments. Operationally the station is divided into two sections : the Russian Orbital Segment assembled by Roscosmos and the US Orbital Segment assembled by NASA, JAXA, ESA and CSA.
- The ISS maintains an orbit with an average altitude of 400 kilometres (250 mi) and circles the Earth in roughly 93 minutes, completing 15.5 orbits per day.

- The ISS programme combines two prior plans to construct crewed Earth-orbiting stations : Space Station Freedom planned by the United States, and the Mir-2 station planned by the Soviet Union. The first ISS module was launched in 1998. Major modules have been launched by Proton and Soyuz rockets and by the Space Shuttle launch system.
- The first long-term residents, Expedition 1, arrived on 2 November, 2000. As of April, 2025, 285 individuals from 23 countries have visited this space station.
- USA launched Marinar-4 in 1964, which photographed some pictures of Mars in July 1965, when it was passing near to Mars.
- On 14 November, 1971 Mariner 9 reached the Mars orbit and became the first space probe to orbit another planet.
- After these mission several campaigns were conducted but twin rover **Spirit** and **Opportunity** and later rover **Curiosity** were highlighted, which were launched by NASA. **Perseverance** is the latest Mars rover of NASA.

Tiangong Space Station :

- Tiangong space station is a permanently crewed space station constructed by China and operated by China Manned Space Agency. Tiangong is a modular design, with modules docked together while in low- Earth orbit, between 340 and 450 km (210 and 280 mi) above the surface.
- It is China's first long-term space station, part of the Tiangong program and the core of the 'Third Step' of the China Manned Space Program.
- The space station aims to provide opportunities for space-based experiments and a platform for building capacity for scientific and technological innovation.
- The construction of the station is based on the experience gained from its precursors, Tiangong-1 and Tiangong-2. The first module, the Tianhe ("Harmony of the Heavens") core module, was launched on 29 April, 2021. This was followed by multiple crewed and uncrewed missions and the addition of two laboratory cabin modules. The first, Wentian ("Quest for the Heavens"), launched on 24 July, 2022; the second, Mengtian ("Dreaming of the Heavens"), launched on 31 October, 2022.

Mars Exploration :

- Mars is the fourth planet from the Sun and the second smallest planet in the Solar System after Mercury.
- It is also known as Red Planet (due to presence of iron oxide on its surface).
- It is a terrestrial planet with a thin atmosphere.
- Its surface reminds the craters of Moon and volcanoes, valleys, deserts, and polar icy peaks of the Earth.
- **Olympus Mons** is a very large shield volcano on the planet Mars. This volcano has a height of approximately 22 km as measured by the Mars Orbiter Laser Altimeter. It is about two and a half times of Mount Everest.
- The largest valley situated on Mars is **Valles Marineris**.

Rover :

- Rover means – Nomadic.
- In celestial reference, rover is such a vehicle, which is capable to rove on or move across the surface of a celestial body.
- Rovers are created to land on any planet or celestial body to find out information and to take samples. They can collect dust, rocks and even take pictures.
- They usually arrive at the planetary surface on a lander style spacecraft.

Spirit :

- Spirit also known as MER-A (Mars Exploration Rover-A) or MER-2, is a robotic rover on Mars, active from 2004 to 2010.
- It was one of two rovers of NASA's on going Mars Exploration Rover Mission.

Opportunity :

- It is also known as MER-B or MER-1 rover.
- It is a robotic rover which was active on Mars from 2004 to June, 2018.
- Before it, under NASA'S Mars Pathfinders Mission– **Sojourner** rover was arrived on Mars (in 1997) which was active only for three months.

Curiosity :

- Curiosity is a car-sized rover designed to explore the crater Gale on Mars (as part of NASA's Mars Science Laboratory Mission) which was landed on Mars in August, 2012.
- This rover is still (As of April, 2025) operational and exploring Mars.

Perseverance :

- It is a car-sized Mars rover designed to explore the Jezero crater on Mars. It was launched on 30 July, 2020 and successfully landed on Mars on 18 February, 2021.
- Following the rover's arrival, NASA named the landing site **Octavia E. Butler Landing**.

- The rover Perseverance also carried the mini-helicopter '**Ingenuity**' to MARS, an experimental aircraft and technology showcase that made the first powered flight on another planet on 19 April, 2021. On 18 January, 2024, it made its 72nd and final flight at Mars, suffering damage on landing to its rotor blades, causing NASA to retire it.
- The rover's goals include identifying ancient Martian environments capable of supporting life, seeking out evidence of former microbial life existing in those environments, collecting rock and soil samples, and testing oxygen production from the Martian atmosphere to prepare for future crewed missions.

Mangalyaan :

- It was a mission of India.
- The Mangalyaan was also called Mars Orbiter Mission (MOM).
- It was a space probe orbiting Mars Since 24 Sept., 2014.
- It was launched on 5 November, 2013 by ISRO.
- Its main aim was to investigate water, methane and life on Mars.
- India made world record by successful installation of Mangalyaan as it is the first nation in the world to do so in its first attempt.
- India became first Asian country by acquiring this achievement.
- In October, 2022 this mission was officially declared as dead.

UAE's Mars Mission 'Hope' :

- UAE's Mars orbiter mission 'Hope' (Arabic : Misbar Al-Amal) was launched on 19 July, 2020 from the Tanegashima Space Centre in Japan with a Japanese rocket.
- The Hope orbiter went into orbit around Mars on 9 February, 2021 and UAE became the first Arab country and the fifth spacefaring entity (after the USA, the Soviet Union, Europe and India) to reach Mars and the second country to successfully enter Mars' orbit on its first try (after India).

Tianwen-1 :

- It is an interplanetary mission by the China National Space Administration (CNSA) to send a robotic spacecraft to Mars, consisting of : an orbiter, two deployable cameras, lander, remote camera, and the **Zhurong** rover.
- The mission was launched on 23 July, 2020 and entered Martian orbit on 10 February, 2021.

- On 14 May 2021, the lander/rover portion of the mission successfully touched down on Mars, making China the third nation (after the Soviet Union and the USA) to make a soft landing on and establish communication from the Martian surface.
- On 22 May 2021, the Zhurong rover drove onto the Martian surface and with the successful deployment of the rover, China became the second nation to accomplish this feat, after the USA.
- In addition, China is the first nation to carry out an orbiting, landing and roving mission on Mars successfully on its maiden attempt.
- Tianwen-1 is also the second mission to capture audio recordings on the Martian surface, after USA's Perseverance rover.
- The Zhurong rover lost contact in 2023 after failing to awake from a scheduled hibernation.

Artemis Accords :

- The Artemis Accords is a series of non-binding bilateral arrangements between the United States government and other world governments that elaborates on the norms expected to be followed in outer space.
- The accords are related to the Artemis program, an American-led effort to return humans to the Moon by 2027, with the ultimate goal of expanding space exploration to Mars and beyond. The program's stated long-term goal is to establish a permanent base on the Moon to facilitate human missions to Mars.
- The Accords were signed on 13 October, 2020 by representatives of the national space agencies of eight countries : Australia, Canada, Italy, Japan, Luxembourg, the United Arab Emirates, the United Kingdom and the United States.
- The Accords remain open for signature indefinitely, as NASA anticipates more nations joining. Additional signatories can choose to directly participate in Artemis program activities, or may agree simply to commit to the principles for responsible exploration of the Moon as set out in the Accords.
- As of 8 April, 2025, fifty-four countries have signed the accords, including twenty-seven in Europe, Ten in Asia (including India and Bangladesh), seven in South America, five in North America, three in Africa and two in Oceania.

International Lunar Research Station :

- The International Lunar Research Station (ILRS) is a planned lunar base currently being led by Russian space agency Roscosmos and the China National Space Administration (CNSA).
- The ILRS will serve as a comprehensive scientific experiment base built on the lunar surface or in lunar orbit that can carry out multi-disciplinary and multi-objective scientific research activities including exploration and utilization, lunar-based observation, basic scientific experiment and technical verification, and long-term autonomous operation.
- On 9 March, 2021 China and Russia signed a Memorandum of Understanding between the Government of the People's Republic of China and the Government of the Russian Federation regarding cooperation for the construction of the International Lunar Research Station. Statements from Roscosmos and CNSA underline that the project will be open to all interested countries and international partners.
- As of April, 2025, eleven more countries have joined this initiative. These are : Venezuela, South Africa, Azerbaijan, Pakistan, Belarus, Egypt, Thailand, Nicaragua, Serbia, Kazakhstan and Senegal.

Indian Lunar Exploration Missions :

I. Chandrayaan -1

- Chandrayaan-1, India's first mission to Moon, was launched successfully on 22 October, 2008 by PSLV-C11 from Satish Dhawan Space Centre SHAR, Sriharikota. The mission included a lunar orbiter and an impactor.
- The spacecraft was orbiting around the Moon at a height of 100 km from the lunar surface for chemical, mineralogical and photo-geologic mapping of the Moon.
- After the successful completion of all the major mission objectives and more than 3400 orbits around the Moon, the mission was concluded when the communication with the spacecraft was lost on 29 August, 2009.

II. Chandrayaan -2

- India's Geosynchronous Satellite Launch Vehicle, GSLV MkIII (LVM3)-M1, successfully launched Chandrayaan-2 spacecraft on July 22, 2019 from Satish Dhawan Space Centre SHAR, Sriharikota.
- Chandrayaan-2 mission comprised an Orbiter, Lander and Rover to explore the unexplored South Pole of the Moon. On August 20, 2019, Chandrayaan-2 was successfully inserted into lunar orbit.

- While orbiting the Moon in a 100 km lunar polar orbit, on September 2, 2019, **Vikram** Lander (with **Pragyan** Rover) was separated from the Orbiter in preparation for landing. Vikram Lander descent was as planned and normal performance was observed upto an altitude of 2.1 km. Subsequently communication from lander to the ground stations was lost on September 7, 2019.
- The Vikram lander carrying the rover crash-landed and was destroyed upon impact with the Moon's surface. Pragyan, being onboard Vikram, was also destroyed. A successful soft landing would have made India the fourth country after the Russia, United States and China to do so.
- The Orbiter of this mission placed in its intended orbit around the Moon will enrich our understanding of the moon's evolution and mapping of the minerals and water molecules in Polar regions, using its eight state-of-the-art scientific instruments. The precise launch and mission management has ensured a long life of almost 7.5 years instead of the planned one year.

III. Chandrayaan -3

- Chandrayaan-3 is a follow-on mission to Chandrayaan-2 to demonstrate end-to-end capability in safe landing and roving on the lunar surface.
- The mission consists of a Vikram lunar lander and a Pragyan lunar rover similar to those launched aboard Chandrayaan-2 in 2019, as well as a propulsion module that carried the spacecraft from Earth orbit to lunar orbit.
- It was launched by GSLV-MkIII (LVM3-M4) from SDSC, SHAR, Sriharikota on 14 July, 2023.
- The spacecraft entered lunar orbit on 5 August, 2023 and became the first lander to touch down near the lunar south pole on 23 August, 2023, making India the fourth country to successfully land on the Moon.
- The propulsion module has Spectro-polarimetry of HAbitable Planet Earth (SHAPE) payload to study the spectral and polarimetric measurements of Earth from the lunar orbit.
- **Lander payloads :** Chandra's Surface Thermophysical Experiment (ChaSTE) to measure the thermal conductivity and temperature; Instrument for Lunar Seismic Activity (ILSA) for measuring the seismicity around the landing site; Langmuir Probe (LP) to estimate the plasma density and its variations. A passive Laser Retroreflector Array from NASA is accommodated for lunar laser ranging studies.

- **Rover payloads :** Alpha Particle X-ray Spectrometer (APXS) and Laser Induced Breakdown Spectroscopy (LIBS) for deriving the elemental composition in the vicinity of landing site. APXS was used to determine the elemental composition (Mg, Al, Si, K, Ca, Ti, Fe) of lunar soil and rocks around the lunar landing site, while LIBS was used for qualitative and quantitative elemental analysis and to derive the chemical and infer mineralogical composition to further our understanding of lunar surface.

IV. Chandrayaan-4 Mission

- In September, 2024, the Union Cabinet chaired by the PM Narendra Modi has approved the Chandrayaan-4 mission.
- Chandrayaan-4 mission, aims at developing and demonstrating key technologies required for returning to Earth after a successful lunar landing. This mission will also focus on collecting samples from the Moon and analyzing them on Earth.
- Chandrayaan-4 is a critical step toward establishing India's capabilities for a human mission to the Moon, planned for 2040. The mission will showcase advanced technologies for docking, undocking, landing and returning safely to Earth, alongside lunar sample collection and analysis.
- The successful soft landing of Chandrayaan-3 demonstrated vital technologies that only a few nations possess. Building on this, Chandrayaan-4 will focus on returning lunar samples safely to Earth, a critical capability in future manned missions.
- The mission, managed by ISRO and expected to be completed within 36 months of approval, will have strong participation from industry and academia. All critical technologies will be developed indigenously, creating high employment potential and technological spin-offs for other sectors.
- The total funding requirement is Rs. 2,104.06 crore, covering spacecraft development, two launch vehicle missions of LVM3, external deep space network support, and special tests for design validation.

LUPEX Mission :

- The Lunar Polar Exploration (LUPEX) Mission, a collaborative effort between India's Indian Space Research Organisation (ISRO) and Japan's Japan Aerospace Exploration Agency (JAXA), is an innovative mission focused on investigating the Moon's south pole.
- This mission represents a notable advancement in lunar research and has significant implications for future space

endeavours, including the potential for a permanent human presence on the Moon.

- Scheduled for launch in the late 2020s, LUPEX will enhance our understanding of critical resources on the Moon's surface, particularly concentrating on water ice deposits.
- In this partnership, JAXA is tasked with creating a sophisticated rover that can traverse and perform research on the Moon's surface, while ISRO is set to provide a reliable lander for deploying the rover.
- Furthermore, the mission will include instruments from NASA and the European Space Agency (ESA), allowing for a wide variety of scientific experiments and data collection.

NISAR Satellite :

- NISAR (NASA-ISRO Synthetic Aperture Radar) satellite is a joint Earth-observing mission between NASA and the Indian Space Research Organisation (ISRO) with the goal to make global measurements of the causes and consequences of land surface changes using advanced radar imaging.
- In 2014, NASA and ISRO signed a partnership to collaborate on and launch NISAR. The mission is expected to launch in 2025.
- NISAR will be the first satellite mission to use two different radar frequencies (L-band and S-band) to measure changes in our planet's surface less than a centimeter across.
- NASA is providing the mission's L-band synthetic aperture radar, a high-rate communication subsystem for science data, GPS receivers, a solid-state recorder and payload data subsystem.
- ISRO is providing the spacecraft bus, the S-band radar, the launch vehicle and associated launch services.

Aditya-L1 Mission :

- On 2 September, 2023, the Polar Satellite Launch Vehicle (PSLV-C57) successfully launched the Aditya-L1 spacecraft, from the Second Launch Pad of Satish Dhawan Space Centre (SDSC), Sriharikota.
- Aditya-L1 is the first Indian space based observatory to study the Sun from a halo orbit around first Sun-Earth Lagrangian point (L1), which is located roughly 1.5 million km from Earth.
- Aditya-L1 carries seven scientific payloads indigenously developed by ISRO and national research laboratories including Indian Institute of Astrophysics (IIA), Bengaluru and Inter University Centre for Astronomy & Astrophysics (IUCAA), Pune.

- On 6 January, 2024, Aditya-L1 was successfully injected on the Halo orbit of Lagrange point 1.
 - The instruments of Aditya-L1 are tuned to observe the solar atmosphere mainly the chromosphere and corona. In-situ instruments will observe the local environment at L1. There are total seven payloads on-board with four of them [Visible Emission Line Coronagraph (VELC), Solar Ultraviolet Imaging Telescope (SUIT), Solar Low Energy X-ray Spectrometer (SoLEXS) and High Energy L1 Orbiting X-ray Spectrometer (HEL1OS)] carrying out remote sensing of the Sun and three of them [Aditya Solar wind Particle Experiment (ASPEX), Plasma Analyser Package For Aditya (PAPA) and Advanced Tri-axial High Resolution Digital Magnetometers] carrying in-situ observation.
 - Aditya-L1 will provide a greater advantage of observing the solar activities and its effect on space weather in real time.
 - The suits of Aditya-L1 payloads are expected to provide most crucial informations to understand the problem of coronal heating, coronal mass ejection, pre-flare and flare activities and their characteristics, dynamics of space weather, propagation of particle and fields etc.
 - On 22 February, 2024, Aditya L1 captured the first-ever image of a X6.3-class solar flare 'kernel' occurring in the photosphere and the chromosphere. The image was taken alongside operations of SUIT, SoLEXS and HEL1OS instruments in the Near-Ultraviolet band. On 16 July, 2024, the VELC instrument was used to measure a large Coronal Mass ejection from the Sun. It also studied an accompanying solar flare and the motion of solar particles within the Sun. Scientist from IIA reported observations of a flareless coronal mass ejection from reading taken by VELC in March, 2025.
 - Marking its first year of science observation, ISRO released the maiden set of the scientific data from the Aditya L1 to the global scientific community on 6 January, 2025 at the ISRO Headquarters in Bengaluru. ISRO presented the second set on 14 February, 2025.
- Gaganyaan :**
- Gaganyaan project envisages demonstration of human spaceflight capability by launching crew of 3 members to an orbit of 400 km for a 3 days mission and bring them back safely to Earth, by landing in Indian sea waters.
 - The project is accomplished through an optimal strategy by considering inhouse expertise, experience of Indian industry, intellectual capabilities of Indian academia and research institutions along with cutting edge technologies available with international agencies.
 - The pre-requisites for Gaganyaan mission include development of many critical technologies including human rated launch vehicle for carrying crew safely to space, Life Support System to provide an Earth like environment to crew in space, crew emergency escape provision and evolving crew management aspects for training, recovery and rehabilitation of crew.
 - Various precursor missions are planned for demonstrating the Technology Preparedness Levels before carrying out the actual Human Space Flight mission. These demonstrator missions include Integrated Air Drop Test (IADT), Pad Abort Test (PAT) and Test Vehicle (TV) flights. Safety and reliability of all systems will be proven in unmanned missions preceding manned mission.
 - LVM3 rocket – The well proven and reliable heavy lift launcher of ISRO, is identified as the launch vehicle for Gaganyaan mission. It consists of solid stage, liquid stage and cryogenic stage. All systems in LVM3 launch vehicle are re-configured to meet human rating requirements and christened Human Rated LVM3. HLVM3 will be capable of launching the Orbital Module to an intended Low Earth Orbit (LEO) of 400 km.
- Bharatiya Antariksh Station :**
- In September, 2024, the Union Cabinet chaired by the PM Narendra Modi has approved the building of first unit of the Bharatiya Antariksh Station by extending the scope of Gaganyaan program.
 - The construction of the first module of the Bharatiya Antariksh Station (BAS-1) marks a significant expansion of the Gaganyaan program. This decision paves the way for missions aimed at developing and validating the technologies required to build and operate the Indian space station. The revised scope of the Gaganyaan program will now include precursor missions for BAS, additional hardware requirements, and the inclusion of an additional uncrewed mission.
 - The Gaganyaan program, initially approved in December, 2018, was designed to launch human spaceflight missions to Low Earth Orbit (LEO) and lay the groundwork for India's long-term ambitions in human space exploration.
 - The expanded vision of the Indian space program during

Amrit Kaal includes establishing an operational Bharatiya Antariksh Station by 2035 and sending an Indian crewed mission to the Moon by 2040. The program will involve eight missions, to be completed by 2028, including the launch of the first unit of BAS-1.

- ISRO will lead the Gaganyaan program, working closely with industry, academia, and other national agencies. Four missions under the current Gaganyaan plan are expected by 2026, followed by four additional missions focused on demonstrating and validating technologies for the space station by December, 2028.
- The revised Gaganyaan program now has a total funding allocation of Rs. 20,193 crore, with an additional Rs. 11,170 crore in newly approved funding.

SpaDeX Mission :

- In a historic achievement, the Space Docking Experiment (SpaDeX) mission's docking operation was successfully completed on 16 January, 2025 marking India's entry into the elite group of nations capable of executing space docking operations.
- With this success, India becomes the fourth country in the world to achieve this technological feat.
- ISRO started the mission on 30 December, 2024, with the successful launch of the SpaDeX satellites (A&B; SDX-01 and SDX-02) using Polar Satellite Launch Vehicle (PSLV)-C60, from the Satish Dhawan Space Centre, Sriharikota.
- This groundbreaking mission aims to showcase India's technological prowess in spacecraft rendezvous, docking, and undocking – a critical capability for future advancements such as satellite servicing, space station operations, and interplanetary exploration.
- On 16 January, 2025, the docking process was executed with exceptional precision. Post-docking, the integrated control of the two satellites as a single object has been successfully achieved, showcasing India's technological expertise.
- Notably, India became the fourth country in the world to achieve this successful space docking after the USA, Russia, and China.
- SpaDeX is a cost-effective technology demonstrator mission designed to showcase in-space docking using two small spacecrafts launched by 62nd PSLV flight. This mission is pivotal for India's future space ambitions, including lunar missions, sample returns, and the development of the Bharatiya Antariksh Station (BAS).

The main goals of the SpaDeX mission are :

- ♦ Develop and demonstrate technology for rendezvous and docking using two small spacecrafts.
- ♦ Showcase controllability in the docked condition.
- ♦ Demonstrate the potential for extending the life of the target spacecraft.
- ♦ Test power transfer between docked spacecraft.
- On 13 March, 2025, ISRO successfully undocked the SpaDeX satellites, a significant achievement demonstrating India's capabilities in rendezvous, docking and undocking operations in a circular orbit, paving the way for future missions like Chandrayaan-4 and a planned space station.
- After successful undocking, the spacecraft separated and will be further used for application missions. During undocking, the spacecraft separated to begin individual payload operations. These payloads will provide high resolution images, natural resource monitoring, vegetation studies and on orbit radiation environment measurements which find numerous applications.
- On 20 April, 2025, ISRO successfully demonstrated the docking of SDX-01 and SDX-02 spacecraft for the second time. On the next day, the electrical power transfer from SDX-02 and SDX-01 and vice versa was also completed successfully.
- Space docking is a critical prerequisite for upcoming space missions, such as lunar exploration and the operation of space stations. By successfully executing this mission, ISRO is laying the foundation for autonomous docking – a vital capability for future missions like Chandrayaan-4. Additionally, the SpaDeX mission will play a key role in supporting India's ambitious space goals, such as the Gaganyaan mission, sending an Indian astronaut to the Moon and setting up the Bharat Antariksh Station.

eLISA Project :

- eLISA (evolved Laser Interferometer Space Antenna) is a proposed space probe to detect and accurately measure gravitational waves.
- It is a space-based gravitational wave observatory building on the success of LISA Pathfinder and LIGO.
- The European Space Agency (ESA) is leading this mission with supports from NASA and an international consortium of scientists.
- eLISA is a spectacular plan of setting into space three spacecrafts, a mother and two daughter spacecraft, which will fly in a triangular formation, trailing the Earth in its orbit around the sun at a distance of over 50 million

km. Each arm of the imaginary equilateral triangle will measure about a million km.

- The distance between the spacecrafts is precisely monitored by laser interferometers to detect a passing gravitational wave. This mission is slated for 2034.

VERITAS and DAVINCI+ :

- 'VERITAS' and 'DAVINCI+' will be NASA's first missions to Venus since the 1990s. VERITAS is tentatively scheduled to launch in December, 2027 and would begin orbiting Venus in July, 2028 (DAVINCI+'s launch is targeted for 2029). These missions are Part of NASA's Discovery Programme.
- The VERITAS (Venus Emissivity, Radio Science, InSAR, Topography, and Spectroscopy) mission will map the surface of the planet, study its geology, and hunt for volcanic activity. VERITAS – which means truth in Latin – will help scientists learn the facts about how Venus became a sulfurous inferno, while Earth evolved to become home to an abundance of life.
- The DAVINCI (Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging) with its atmospheric probe, called Plus, will study the dense atmosphere of Venus to understand the trigger and evolution of the runaway greenhouse effect active on the planet.

Venus Orbiter Mission (VOM) :

- In September, 2024, the Union Cabinet chaired by the PM Narendra Modi has approved the development of Venus Orbiter Mission (VOM).
- The ambitious Venus Orbiter Mission (VOM) marks a major step forward in India's planetary exploration efforts beyond the Moon and Mars. Venus, the planet closest to Earth and believed to have formed under conditions similar to our own, presents a unique opportunity to study how planetary environments can evolve in dramatically different ways. This mission aligns with the government's long-term vision of advancing space exploration capabilities.
- The Venus Orbiter Mission, led by the Department of Space, will place a scientific spacecraft in orbit around Venus. The mission aims to improve our understanding of the planet's surface, subsurface, atmospheric processes, and the Sun's influence on its atmosphere. Investigating the causes behind Venus's transformation from a potentially habitable planet to its current state will provide crucial insights into the evolutionary paths of both Venus and Earth.

- ISRO will oversee the development and launch of the spacecraft, following its well-established project management practices. The data collected from the mission will be shared with the scientific community, fostering further research.
- The mission is expected for launch in March, 2028, offering significant scientific insights and various outcomes, particularly in answering unresolved questions about Venus.
- The total approved budget for the Venus Orbiter Mission is Rs. 1,236 crore, with Rs. 824 crore allocated to spacecraft development. The cost includes development and realization of the spacecraft including its specific payloads and technology elements, global ground station support cost for navigation and network as well as the cost of launch vehicle.

Question Bank

1. Who was the father of Indian Space Programme?

- (a) Dr. Vikram Ambalal Sarabhai
- (b) Dr. Satish Dhawan
- (c) Dr. Homi J. Bhabha
- (d) Dr. Krishnaswami Kasturirangan

M.P.P.C.S. (Pre) 2019

Ans. (a)

Dr. Vikram Ambalal Sarabhai was an Indian physicist and astronomer who initiated space research in India. He is considered as the Father of the Indian Space Programme.

2. Who amongst the following is considered as father of India's space programme?

- (a) Nambi Narayanan
- (b) G. Madhavan Nair
- (c) Vikram Sarabhai
- (d) A.P.J. Abdul Kalam

Jharkhand P.C.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

3. The First Satellite of India sent to space, was :

- (a) Bhaskar
- (b) Rohini
- (c) Aryabhata
- (d) Apple

U.P. Lower Sub. (Pre) 2009

M.P.P.C.S. (Pre) 1992

U.P.P.C.S. (Mains) 2006

Ans. (c)

Aryabhata was the first satellite launched by India. It was named after the great Indian astronomer of the same name. Aryabhata weighed 360 kg and was launched by the Soviet Union on April 19, 1975 from Kapustin Yar using a Cosmos-3M launch vehicle.

4. **India entered space age by launching –**
(a) Bhaskara I (b) Aryabhata
(c) Apple (d) Agni
R.A.S./R.T.S.(Pre) 2013

Ans. (b)

See the explanation of above question.

5. **Aryabhata was launched on :**
(a) April 19th, 1975 (b) December 6th, 1957
(c) January 31, 1958 (d) October 2nd, 1980
M.P.P.C.S. (Pre) 2008

Ans. (a)

See the explanation of above question.

6. **The name of scientist who designed India's first space satellite 'Aryabhata' and died recently is :**
(a) Prof. U.R. Rao (b) Prof. Yashpal
(c) Prof. C.N. R. Rao (d) Dr. A. S. Kiran Kumar
U.P.P.S.C. (GIC) 2017

Ans. (a)

Prof. U.R. Rao was the scientist who designed India's first space satellite 'Aryabhata'. Rao undertook the responsibility for the establishment of satellite technology in India in 1972. Under his guidance, beginning with the first Indian satellite 'Aryabhata' in 1975, over 18 satellites were designed, fabricated and launched. Rao also served as the Chairman of ISRO for 10 years from 1984-1994. He passed away at the age of 85 on 24 July, 2017.

7. **Which was the first animal to travel to space?**
(a) A mouse (b) A dog
(c) A cockroach (d) A mosquito
U.P.P.S.C. (GIC) 2010

Ans. (b)

Laika was a Soviet space dog who became the first animal to orbit the earth in space. Laika, a stray dog from the streets of Moscow, was selected to be the occupant of the Soviet spacecraft Sputnik 2 that was launched into outer space on November 3, 1957.

8. **Which of the following names of animals was sent to space on November 3rd, 1957?**
(a) Otolith (b) Laika
(c) Baker (d) Belka
M.P.P.C.S. (Pre) 2010

Ans. (b)

On November 3rd, 1957 a dog, named 'Laika' was sent to space- through Sputnik-2 spacecraft. On 28 May, 1959 monkeys named 'Able' and 'Baker' were sent to space. On August 19, 1960 a dog named 'Belka' was sent to space through Sputnik-5 spacecraft. On 28 January, 2013 Iran sent a monkey into space by Pishgam rocket.

9. **Who is the first astronaut in space?**
(a) Yuri Gagarin (b) Alan Shepard
(c) S. Titov (d) John Yung
M.P.P.C.S. (Pre) 2010

Ans. (a)

On 12 April, 1961 astronaut Yuri Gagarin became the first human in space by Vostok-1 Russian spacecraft. Astronaut Alan Shepard became the first American in space less than a month later.

10. **Who among the following is world's first woman cosmonaut :**
(a) Bachendri Pal (b) Junko Tabei
(c) Valentina Tereshkova (d) Sally Ride
U.P. Lower Sub. (Spl.) (Pre) 2004
U.P.P.C.S. (Pre) 2001

Ans. (c)

Russian astronaut Yuri Gagarin was the first male astronaut in space while Valentina Tereshkova (Russian cosmonaut) became the first woman to fly to space, when she launched on the Vostok-6 mission on 16 June, 1963. She spent almost three days in space and orbited Earth 48 times.

11. **The first man who placed his foot on the Moon is :**
(a) Leonov (b) Neil Armstrong
(c) Michael Collins (d) James Van Allen
(e) None of the above / More than one of the above
66th B.P.S.C. (Pre) 2020

Ans. (b)

Apollo-11 was the spaceflight that landed the lunar module 'Eagle' with two people on the Moon on 20th July, 1969. The NASA astronaut and mission commander Neil Armstrong was the first man who placed his foot on the Moon and his fellow astronaut pilot Buzz Aldrin joined him at the Moon's surface after a gap of about 19 minutes.

12. **Cosmonaut Oleg Kononenko who has broken the world record for most cumulative time spent in the space, belongs to :**
(a) Ukraine (b) Japan
(c) USA (d) Russia
Uttarakhand P.C.S. (Pre) 2024

Ans. (d)

Oleg Kononenko is a Russian cosmonaut. Kononenko has accumulated 1110.6 days, in space over the course of five long-duration missions on the International Space Station (ISS). On 4 February, 2024, he broke the record of the most cumulative time spent in space previously held by Gennady Padalka of Russia at 878.5 days.

13. Who is the record holder for single longest space flight by a 'NASA' astronaut abroad the International Space Station?

- (a) Pyotr Dubrov (b) Mark Vande Hei
(c) Anton Sheaplerov (d) None of the above

U.P. P.C.S. (Pre) 2022

Ans. (*)

At present, NASA astronaut Francisco (Frank) Rubio holds the longest single spaceflight record by an American at 370.9 days. He returned to Earth on 27 September, 2023. Earlier, NASA astronaut Mark Vande Hei completed the then record single mission by an American aboard the International Space Station of 355.2 days on March 30, 2022. Soviet and Russian cosmonaut Valeri Polyakov holds the longest single human spaceflight record at 437.7 days.

14. Tim Peake is known as which of the following?

- (a) Writer (b) Physicist
(c) Astronaut (d) Politician

U.P. Lower Sub. (Pre) 2015

Ans. (c)

Tim Peake is a European Space Agency (ESA) astronaut of British nationality and a former International Space Station (ISS) crew member. Tim has a background as a test pilot and a British Army Air Corps officer.

15. Which one of the following pairs is not correctly matched?

- (a) Cosmic Background Explorer (COBE) – Satellite programme
(b) Falcon – Under-sea cable system
(c) Discovery – Space shuttle
(d) Atlantis – Space station

I.A.S. (Pre) 2007

Ans. (d)

Columbia, Challenger, Discovery and Atlantis—operations of all these space shuttles have now been closed by NASA. It may be noted that the space shuttles may be launched like rockets, revolves around the Earth like satellite and again land on the Earth like an aircraft.

COBE was a satellite dedicated to cosmology, which operated from 1989 to 1993. Its goals were to investigate the cosmic microwave background radiation of the universe and provide measurements that would help shape our understanding of the cosmos. Falcon is the largest submarine fibre optic cable network laid by then Reliance communications. It was inaugurated on 5th September, 2006. Presently, this system connects India with four continents and eleven countries.

16. Match List-I with List-II and select the correct answer by using the codes given below the lists :

List-I

- A. Curiosity Rover**
B. Messenger
C. Rustam-1
D. Akash-2

List-II

- 1. Tablet**
2. Unmanned Aerial vehicle of DRDO
3. Mars probe space vehicle of U.S.A.
4. Mercury probe space vehicle of NASA

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 4 | 2 | 1 |
| (b) | 4 | 3 | 2 | 1 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 1 | 2 | 3 | 4 |

U.P.P.C.S.(Pre) 2013

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (a)

Curiosity: Curiosity is a car-sized robotic rover exploring Gale Crater on Mars as part of NASA's Mars Science Laboratory mission. It was launched from Cape Canaveral on 26 November, 2011.

MESSENGER : Messenger was a robotic NASA spacecraft launched in August, 2004 to study Mercury's chemical composition, geology and magnetic field.

RUSTAM-I : Rustam is a medium Altitude Long Endurance unmanned combat air vehicle being developed by Defence Research and Development Organization (DRDO).

Aakash-II : The new version of the world's cheapest tablet.

17. Which one of the following is NASA's last Space Shuttle?

- (a) Atlantis (b) Columbia
(c) Discovery (d) Endeavor

U.P. Lower Sub. (Pre) 2009

Ans. (a)

Atlantis was the last and final space shuttle of NASA. It was launched on 8 July, 2011 and landed on 21 July, 2011. Columbia space shuttle was disintegrated during re-entry in earth's atmosphere on 1 February, 2003. The last Mission of Discovery space shuttle was held between 24 February to 9 March, 2011. On 16 May, 2011 the Endeavor was launched for its last space mission and it successfully landed on the earth on 1 June, 2011.

18. 'DAVINCI+' and 'VERITAS' – two new mission announced by NASA, are related to :

- (a) Moon (b) Venus
(c) Mars (d) Jupiter

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

'VERITAS' and 'DAVINCI+' will be NASA's first missions to Venus since the 1990s. VERITAS is tentatively scheduled to launch in December 2027, and would begin orbiting Venus in July 2028 (DAVINCI+'s launch is targeted for 2029). These missions are Part of NASA's Discovery Programme. The VERITAS (Venus Emissivity, Radio Science, InSAR, Topography, and Spectroscopy) mission will map the surface of the planet, study its geology, and hunt for volcanic activity. VERITAS – which means truth in Latin – will help scientists learn the facts about how Venus became a sulfurous inferno, while Earth evolved to become home to an abundance of life. The DAVINCI (Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging) with its atmospheric probe, called Plus, will study the dense atmosphere of Venus to understand the trigger and evolution of the runaway greenhouse effect active on the planet.

19. Match List-I with List-II :

List-I (Space Mission)	List-II (Exploration)
A. Cassini-Huygens	1. Jupiter
B. Juno	2. Saturn and its rings
C. Artemis	3. Venus
D. VERITAS	4. Human Space-flight – Moon to Mars

Select the correct answer using the codes given below.

A	B	C	D
(a) 2	1	4	3
(b) 3	1	4	2
(c) 2	3	4	1
(d) 3	1	2	4

69th B.P.S.C. (Pre) 2023

Ans. (a)

Cassini-Huygens was a space research mission by NASA, ESA and ISA to send space probe to study the planet Saturn and its system. Juno is a NASA space probe orbiting the planet Jupiter. The Artemis program is a Moon exploration program that is led by the NASA. The program's stated long-term goal is to establish a permanent human base on the Moon to facilitate human missions to Mars. VERITAS mission is for studies of the surface and interior of Venus with a powerful new generation of scientific tools. Hence, (a) is the correct option.

20. Which of the following pairs is/are correctly matched?

Spacecraft	Purpose
1. Cassini-Huygens	: Orbiting the Venus and transmitting data to the Earth
2. Messenger	: Mapping and investigating the Mercury
3. Voyager 1 and 2	: Exploring the outer Solar System

Select the correct answer using the code given below.

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2014

Ans. (b)

Cassini-Huygens was an unmanned spacecraft sent to the planet Saturn. So, pair 1 is not correctly matched. Messenger was a NASA's robotic spacecraft which orbited the planet Mercury between 2011 and 2015. The Voyager program is a continuing American scientific program that employs two space probes, Voyager1 and Voyager 2 to study the outer Solar System. These space probes were launched by NASA in 1977. Clearly, pair 2 and 3 are correctly matched.

21. NASA's 'Cassini' spacecraft is on mission for which planet?

- (a) Jupiter (b) Mars
(c) Saturn (d) Venus

Uttarakhand P.C.S. (Pre) 2016

Ans. (c)

See the explanation of above question.

22. Which one of the following is a spacecraft?

- (a) Apophis (b) Cassini
(c) Spitzer (d) TechSar

I.A.S. (Pre) 2008

Ans. (b)

Cassini was a spacecraft developed by NASA to study ringed planet (Saturn) and its domain. TechSar is an Israeli reconnaissance satellite launched by India's launching vehicle PSLV. Apophis is an asteroid and Spitzer was a space infrared telescope developed by NASA.

23. "The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft." The experiment in question refers to :

- (a) Voyager-2 (b) New Horizons

- (c) LISA Pathfinder (d) Evolved LISA

I.A.S. (Pre) 2020

Ans. (d)

The experiment in question refers to evolved Laser Interferometer Space Antenna (eLISA) which is a proposed space probe to detect and accurately measure gravitational waves. It is a space-based gravitational wave observatory building on the success of LISA Pathfinder and LIGO. The European Space Agency (ESA) is leading this mission with supports from NASA and an international consortium of scientists. eLISA is a spectacular plan of setting into space three spacecrafts, a mother and two daughter spacecraft, which will fly in a triangular formation, trailing the Earth in its orbit around the sun at a distance of over 50 million km. Each arm of the imaginary equilateral triangle will measure about a million km. The distance between the spacecrafts will be precisely monitored by laser interferometers to detect a passing gravitational wave. This mission is slated for 2035.

24. What is the purpose of 'evolved Laser Interferometer Space Antenna (eLISA)' project?

- (a) To detect neutrinos
(b) To detect gravitational waves
(c) To detect the effectiveness of missile defence system
(d) To study the effect of solar flares on our communication systems

I.A.S. (Pre) 2017

Ans. (b)

See the explanation of above question.

25. Scientists of Britain have built 'Gravity Tractor'. Identify the same from the following.

- (a) A double storeyed 'Jugaad' type tractor trolley to be used in public transport.
(b) A tractor with a mechanism to sweep public roads and is economical to run.
(c) A bullet train runs on the Principle of Earth's 'gravity' with the help of powerful magnets.
(d) A spacecraft with a mechanism which would check asteroids from hitting the Earth.

U.P.P.C.S. (Pre) 2009

Ans. (d)

In 2009, British scientists announced that they have designed a special spacecraft that can save the Earth from a catastrophic collision. The 10 ton spacecraft named 'Gravity Tractor' would be deployed to intercept an asteroid en route to the earth and has the ability to fly 160 ft alongside it. The idea of a gravity tractor was put forward in 2005 by Edward T. Lu and Stanley G. Love.

26. Recently, scientists observed the merger of giant 'black holes' billions of light-years away from the Earth. What is the significance of this observation?

- (a) 'Higgs boson particles' were detected.
(b) 'Gravitational waves' were detected.
(c) Possibility of inter-galactic space travel through 'wormhole' was confirmed.
(d) It enabled the scientists to understand 'singularity'.

I.A.S. (Pre) 2019

Ans. (b)

Recently, scientists observed that billions of light years away, two black holes have collided to create a larger one-the biggest black hole merger yet detected. It has a mass more than 80 times that of the Sun. The significance of this observation is that the gravitational waves were detected. The resulting energy injected into the fabric of spacetime was also record breaking, with five Sun's worth of mass released in the form of gravitational waves as the two holes spiralled in towards each other. Such titanic amounts of energy meant that the signal was still detectable by the time it reached gravitational wave detectors on Earth.

27. Gravitational waves were seen for the first time in :

- (a) September 2014 (b) September 2015
(c) August 2017 (d) October 2017

U.P. P.C.S. (Mains) 2017

Ans. (b)

The first direct observation of gravitational waves was made on 14 September, 2015 and was announced by the LIGO (Laser Interferometer Gravitational Wave Observatory) on 11 February, 2016.

28. Which of the following is/are cited by the scientists as evidence/evidences for the continued expansion of the universe?

1. Detection of microwaves in space.
2. Observation of redshift phenomenon in space.
3. Movement of asteroids in space.
4. Occurrence of supernova explosions in space.

Select the correct answer using the codes given below:

- (a) 1 and 2
(b) Only 2
(c) 1, 3 and 4
(d) None of the above can be cited as evidence

I.A.S. (Pre) 2012

Ans. (a)

Detection of microwaves in space and observation of redshift phenomenon in space are cited by the scientists as evidence for the continued expansion of the universe. In 1998 two research team presented their findings, first headed by Saul Perlmutter and Adam Riess and another by Brian Schmidt. Both of these teams explored that the universe which came into existence after the event of big-bang about 14 billion years ago, is expanding continuously. These two research teams found that over 50 distant supernovae whose light became more weaker than expected, is the sign that the expansion of the universe is accelerating. For their vast contribution in Physics, the Royal Swedish Academy of Science had decided to award them with the Nobel Prize in Physics for 2011.

29. The concept of expanding universe is based on –

- (a) Doppler effect (b) Stark effect
(c) Zeeman effect (d) Raman effect

U.P.P.C.S. (Mains) 2010

Ans. (a)

The concept of expanding universe is based on Doppler effect. It was named after Christian Doppler, who first came up with the idea in 1942. Edwin Hubble used the Doppler Effect to determine that universe is expanding. Hubble found that the light from distant galaxies was shifted towards lower frequencies, to the red end of the spectrum. This is known as a red Doppler shift or a red-shift.

30. Consider the following phenomena :

1. Light is affected by gravity.
2. The Universe is constantly expanding.
3. Matter warps its surrounding space-time.

Which of the above is/are the prediction/predictions of Albert Einstein's General Theory of Relativity, often discussed in media?

- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (d)

When light travels near a huge celestial body like neutron star or black hole it bends towards the celestial body. Hence, statement 1 is correct. As per the prediction of Albert Einstein, the astronomers have found that the universe is expanding continuously and a heavenly body with huge mass tends to bend (warp) the space-time. This is known as warping of the space-time. This is why light seems to bend when it passes near a neutron star or black hole. Hence, statement 2 and 3 are also correct.

31. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of :

- (a) Observation and understanding of the Universe
(b) Study of the solar and the lunar eclipses
(c) Placing satellites in the orbit of the Earth
(d) Origin and evolution of living organisms on the Earth

I.A.S. (Pre) 2017

Ans. (a)

The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of observation and understanding of the Universe.

In astrophysics, an event horizon is a boundary beyond which events cannot affect an observer. In an expanding universe, the speed of expansion reaches – and even exceeds – the speed of light, preventing signals from traveling to some regions. With reference to black holes, event horizon is the boundary defining the region of space around a black hole from which nothing (not even light) can escape.

Singularities are regions of space where the density of matter, or the curvature of spacetime, becomes infinite. In such locales, the standard concepts of space and time cease to have any meaning. Singularities are predicted to occur in all black holes and also in certain models of the Universe. String theory is a concept in theoretical physics that states the universe is constructed by tiny vibrating strings, smaller than the smallest subatomic particles. According to this theory, as the strings vibrate, twist and fold, they produce effects in many, tiny dimensions that humans interpret as everything from particle physics to large-scale phenomena like gravity. The Standard Model of particle physics is the theory describing three of the four known fundamental forces (electromagnetic force, weak nuclear force, and the strong nuclear force – excluding gravity) in the universe and classifying all known elementary particles.

32. Black-hole is :

- (a) a flight recorder in aeroplane
(b) a spot on the sun
(c) a place in Antarctica
(d) a collapsed star

U.P.P.C.S. (Pre) 2019

Ans. (d)

Black hole is actually a collapsed star. Black holes of stellar mass are expected to form when very massive stars collapse at the end of their life cycle. After a black hole has formed, it can continue to grow by absorbing mass from its surroundings. By absorbing other stars and merging with other black holes, supermassive black holes of millions of solar masses may form.

A black hole is a place in space where gravity pulls so much that even light cannot get out. The gravity is so strong because matter has been squeezed into a tiny space.

33. Assertion (A): Black hole is such a celestial existence which cannot be seen by telescope.

Reason (R) : The gravitational field of black hole is much strong that even light does not escape from it.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (a)

A black hole is an object which is so compact that its gravitational force is strong enough to prevent light or anything else from escaping. By this reason, one can't see it by telescope.

34. Black Hole –

- (a) Does not emit any radiation.
- (b) Converts UV radiation to infrared.
- (c) Absorbs all radiations that fall on it.
- (d) Is a imaginary concept.

Uttarakhand P.C.S. (Pre) 2007

Ans. (c)

Black Hole absorbs all radiations that fall on it due to its high gravity.

35. Which of the following has highly elongated elliptical orbit ?

- (a) Asteroid
- (b) Comet
- (c) Meteorite
- (d) Meteor

70th B.P.S.C. (Pre) 2024

Ans. (b)

Comets are known for their highly elongated, elliptical orbits around the Sun, meaning they spend a significant amount of time far from the Sun and then rapidly approach it, before moving back out into the solar system's outer regions. Comets follow paths that are very different from the nearly circular orbits of planets. Their orbits are stretched out, with a long, thin shape, rather than a more circular or oval shape. A comet's orbit has a point closest to the Sun (perihelion) and a point farthest from the Sun (aphelion). During perihelion, the comet becomes visible as it heats up and its icy material

vaporizes, forming a tail. Asteroids generally have more circular or slightly elliptical orbits, primarily located in the asteroid belt between Mars and Jupiter.

36. The tail of a comet always points –

- (a) Away from the Sun
- (b) Towards the Sun
- (c) Towards North-East
- (d) Towards South-East

U.P.P.C.S. (Mains) 2011

Ans. (a)

Comets are the small celestial objects made of ice, gas, dust, and a small amount of organic material that orbit our Sun. Radiation from the Sun pushes dust particles away from the coma (nebulous envelope around the nucleus of a comet), forming a dust tail, while charged particles from the Sun convert some of the comet's gases into ions, forming an ion tail. Since comet tails are shaped by sunlight and the solar wind, they always point away from the Sun.

37. The bright outer layer of the Sun is known as :

- (a) Photosphere
- (b) Ionosphere
- (c) Mesosphere
- (d) Sunspots

Uttarakhand P.C.S. (Pre) 2024

Ans. (a)

The bright outer layer of the Sun is known as the photosphere. The word photosphere means 'light sphere' – which is apt because this is the layer that emits the most visible light. It's what we see from Earth with our eyes. Although we call it the Sun's surface, the photosphere is actually the first outer layer (from inside) of the solar atmosphere. The other two are, the chromosphere, and the corona.

38. Highly dense stars are called –

- (a) Supernova
- (b) Neutron stars
- (c) Binary stars
- (d) Asteroid

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

The highly dense stars are called Neutron stars.

39. In the universe, what are pulsars?

- (a) A group of stars
- (b) Rotating neutron stars
- (c) Explosion of a star
- (d) Radio waves emitted by a star

69th B.P.S.C. (Pre) 2023

Ans. (b)

Most neutron stars are observed as pulsars. Pulsars are rotating neutron stars observed to have pulses of radiation at very regular intervals that typically range from milliseconds to seconds. Pulsars have very strong magnetic fields which funnel jets of particles out along the two magnetic poles.

40. Consider the following pairs :

Objects in space	Description
1. Cepheids	: Giant clouds of dust and gas in space
2. Nebulae	: Stars which brighten and dim periodically
3. Pulsars	: Neutron stars that are formed when massive stars run out of fuel and collapse

How many of the above pairs are correctly matched?

- (a) Only one (b) Only two
(c) All three (d) None

I.A.S. (Pre) 2023

Ans. (a)

Cepheids, also called Cepheid Variables, are stars which brighten and dim periodically. This behaviour allows them to be used as cosmic yardsticks out to distances of a few tens of millions of light-years. Hence, pair 1 is not correctly matched. Nebulae are interstellar clouds of gas and dust. Many nebulae are formed from the remnants of dying stars. Nebulae are often also regions where new stars are born. Nowadays, the term 'nebula' refers to an interstellar cloud of dust and gas. Hence, pair 2 is not correctly matched.

Pulsars are rotating neutron stars observed to have pulses of radiation at very regular intervals that typically range from milliseconds to seconds. Neutron stars are formed when massive stars run out of fuel and collapse. Most neutron stars are observed as pulsars. Hence, pair 3 is correctly matched.

41. Consider the following statements:

Statement-I: Giant stars live much longer than dwarf stars.

Statement-II: Compared to dwarf stars, giant stars have a greater rate of nuclear reactions.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
(c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

I.A.S. (Pre) 2024

Ans. (d)

Giant stars have masses from eight times to as much as 100 times that of the mass of our sun. These massive stars have hotter and denser cores than dwarf stars (stars that are smaller than five times the mass of our sun). Therefore, giant stars have a greater rate of the nuclear reactions that light up stars and with their increased nuclear reaction rates,

they are more luminous. This immense output of energy is not without consequence. Giant stars actually have shorter lifetimes compared to dwarf stars. Giant stars use up the hydrogen fuel in their core faster, despite starting out with much more of it, meaning they live much shorter lives than dwarf stars. An eight solar mass star will live less than 100 million years. At 10-15 solar masses, the lifetime of the star drops to only 10-20 million years. The most massive giant stars are believed to live no more than a few million years. These lifetimes are a blink of an eye when compared to the 10 billion-year lifetime of the sun. A giant star will also end its life via a supernova explosion, leaving behind a strange object such as a neutron star or an even more bizarre black hole. Hence, Statement-I is incorrect, but Statement-II is correct.

42. Electrically charged particles from space travelling at speeds of several hundred km/sec can severely harm living beings if they reach the surface of the Earth. What prevents them from reaching the surface of the Earth?

- (a) The Earth's magnetic field diverts them towards its poles.
(b) Ozone layer around the Earth reflects them back to outer space.
(c) Moisture in the upper layers of atmosphere prevents them from reaching the surface of the Earth.
(d) None of the statement (a), (b) and (c) given above is correct.

I.A.S. (Pre) 2012

Ans. (a)

Electrically charged particles from space travelling at speed of several hundred km/sec cannot reach the surface of the Earth because the Earth's magnetic field diverts them towards its poles. Its best example is 'Aurora'.

43. If a major solar storm (solar flare) reaches the Earth, which of the following are the possible effects on the Earth?

1. GPS and navigation systems could fail.
2. Tsunamis could occur at equatorial regions.
3. Power grids could be damaged.
4. Intense auroras could occur over much of the Earth.
5. Forest fires could take place over much of the planet.
6. Orbits of the satellites could be disturbed.
7. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

Select the correct answer using the code given below:

- (a) 1, 2, 4 and 5 only (b) 2, 3, 5, 6 and 7 only
(c) 1, 3, 4, 6 and 7 only (d) 1, 2, 3, 4, 5, 6 and 7

I.A.S. (Pre) 2022

Ans. (c)

Solar flares are brief and powerful eruptions of high-energy radiation from the Sun's surface, associated with sunspots. The phenomenon is known to cause radio and magnetic disturbances on Earth. Massive solar flares can disrupt power grids, interfere with high-frequency airline and military communications, disrupt Global Positioning System and navigation signals, interrupt civilian communications, and pose risks to spacecraft and astronauts. Solar flares could interrupt shortwave radio communication of the aircraft flying over polar regions.

Solar flares can temporarily alter the upper atmosphere creating disruptions with signal transmission from, say, a GPS satellite to Earth causing it to be off by many yards. In case of coronal mass ejection or CME, these solar explosions propel bursts of particles and electromagnetic fluctuations into Earth's atmosphere. Those fluctuations could induce electric fluctuations at ground level that could blow out transformers in power grids. CME's particles can also collide with crucial electronics onboard a satellite and disrupt its systems.

During an extreme solar flare event, the auroras (usually which are only visible in the far northern and southern polar regions) could become very intense and could be visible in places that usually never see these elusive night lights. Occurrence of tsunami and forest fires are not associated with the solar flares. Hence, option (c) is the correct answer.

44. Assertion (A): A space shuttle loses radio contact with ground station for a while during landing on the Earth.

Reason (R) : As a space shuttle enters Earth's atmosphere, the shuttle's surface temperature increases which ionize the surrounding air.

In the context of the above statements, which of the following is correct :

- (a) Both (A) and (R) are true, but (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.U.D.A./L.D.A. (Pre) 2002

U.P.P.C.S. (Pre) 2003

Ans. (a)

When a spacecraft re-entered in the Earth's atmosphere, it experiences air drag creating friction. This results in an increase of shuttle's temperature and ionizes the surrounding air. These ionized particles disturbs the radio signals and space shuttle loses radio contact with ground station for a while during landing on the Earth.

45. Consider the following statements :

1. **A geostationary satellite is at an approximate height of 10,000 km.**
2. **FM transmission of music is of very good quality because the atmospheric or man-made noises which are generally frequency variations can do little harm.**

Which of the statements given above is/are correct ?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2005

Ans. (b)

Geostationary or communication satellite is located in space about 36,000 km above the equator. A single geostationary satellite is on a line of sight with about 40 percent of the Earth's surface. Three such satellites each separated by 120 degrees of longitude can provide coverage of the entire planet. A major advantage of FM (Frequency Modulation) in a communication system is the possibility of improved signal-to-noise ratio (SNR or S/N) compared to AM (Amplitude Modulation). Additional techniques, such as pre-emphasis of higher audio frequencies with the corresponding de-emphasis in the receiver, one generally used to improve overall SNR in FM circuits. Since FM signals have constant amplitude, FM receivers normally have limiters that remove AM noise, further improving SNR. Hence, statement 1 is incorrect while statement 2 is correct.

46. The satellite used for telecommunication relay are kept in a geostationary orbit. A satellite is said to be in such an orbit when

1. **The orbit is geosynchronous.**
2. **The orbit is circular.**
3. **The orbit lies in the plane of the Earth's equator.**
4. **The orbit is at an altitude of 22,236 km.**

Select the correct answer using the codes given below

- (a) 1, 2 and 3 (b) 1, 3 and 4
(c) 2 and 4 (d) 1, 2, 3, and 4

I.A.S. (Pre) 2011

Ans. (a)

A geostationary orbit is a circular orbit, about 35,786 kilometres (22,236 mi.) above the Earth's equator. Sometimes it is also known as Clark's orbit. A satellite in a geostationary orbit appears to remain in the same spot in the sky all the time. It is also referred as a geosynchronous equatorial orbit (GEO).

47. What is the geostatic position ?

- (a) When the Earth would stop moving around the Sun.
- (b) When an object lies in between the Moon and the Earth
- (c) When the object moves around the Earth with the velocity commensurate with the velocity of the Earth such that it maintain a fixed position in relation to some place on the Earth.
- (d) When an object occupies a fixed distance from the Earth.

U.P.P.C.S. (Mains) 2005

Ans. (c)

When the object moves around the Earth with the velocity commensurate with the velocity of the Earth such that it maintain a fixed position (fixed height) in relation to some place on the Earth, it is called geostatic position. All the geostationary satellites works on the same principle.

48. Geo-stationary satellite revolves at :

- (a) Any height
- (b) At fixed height
- (c) Height above pole
- (d) Height which depends upon its mass

Chhattisgarh P.C.S. (Pre) 2018

Ans. (b)

See the explanation of above question.

49. In which atmospheric layer, the communication satellites are located ?

- (a) Stratosphere
- (b) Ionosphere
- (c) Troposphere
- (d) Exosphere

U.P.P.C.S. (Mains) 2004

Ans. (d)

The exosphere is the outer most layer of Earth's atmosphere. Air in the Exosphere is extremely thin in many ways. It is almost the same as the airless void of outer space and in this, the communication satellites are located.

50. A geostationary satellite has a period of :

- (a) 6 hours
- (b) 12 hours
- (c) 18 hours
- (d) 24 hours

U.P.P.C.S. (Mains) 2010

Ans. (d)

The geostationary satellite has a same orbital period as the earth's rotation period i.e. 24 hours.

51. The time period of revolution of a geostationary satellite is :

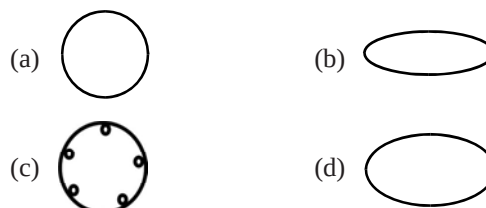
- (a) 12 hours
- (b) 24 hours
- (c) 48 hours
- (d) One year

**U.P. R.O./A.R.O. (Mains) 2016
Uttarakhand P.C.S. (Pre) 2016**

Ans. (b)

See the explanation of above question.

52. Which of the following diagram represents the most accurate path of geostationary satellite in space?



I.A.S. (Pre) 1993

Ans. (a)

Geostationary satellite has a circular path. Therefore option (a) represents the most accurate path of geostationary satellite in space.

53. NASA-U.S. space agency, has launched a telescope named Kepler to find :

- (a) Distant stars
- (b) Distant planets
- (c) Distant satellites
- (d) Earth-like Planets

U.P.P.C.S. (Mains) 2008

Ans. (d)

Kepler was a space telescope launched by NASA to discover Earth-like planets orbiting other stars. It was launched in 2009. The initial planned lifetime for this was 3.5 years but it was expanded. After 9 years of operation, NASA announced its retirement on 30 October, 2018. Kepler discovered more than 2600 'exoplanets'-including many that are promising places for life to exist.

54. Who is the scientist in whose honour the 'Chandra X-ray telescope' has been named –

- (a) Chandrashekhar Venkat Raman
- (b) Jagdish Chandra Bose
- (c) Prafulla Chandra Roy
- (d) Subramanyan Chandrashekhar

I.A.S. (Pre) 2001

Ans. (d)

Chandra X-ray telescope is a space observatory launched by Columbia Space Shuttle on 23 July, 1999. The telescope is named after the Indian-American astrophysicist Subrahmanyan Chandrasekhar.

55. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct?

1. Other than USA and Russia, India is the only country to have launched a similar observatory into space.
2. Astrosat is a 2000 kg satellite placed in an orbit at 1650 km above the surface of the Earth.

Select the correct answer using the code given below :

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2016

Ans. (d)

Astrosat is India's first dedicated multiwavelength space observatory. It was launched on a PSLV-XL on 28 September 2015. This scientific satellite mission endeavours for a more detailed understanding of our universe. One of the unique features of Astrosat mission is that it enables the simultaneous multiwavelength observations of various astronomical objects with a single satellite. Astrosat with a lift-off mass of about 1515 kg was launched into around 650 km orbit at an angle of 6° to the equator by PSLV-C30. Besides NASA, space agencies of the European Union, Japan and Russia have launched similar facilities into space. Hence, both the statements are not correct.

56. Which is the biggest telescope of the world?

- (a) Galileo (b) IceCube
(c) Discovery (d) Challenger

U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

Ans. (b)

The IceCube telescope which was completed in December, 2010 is the biggest neutrino telescope of the world. It was designed by the University of Wisconsin.

57. The world's highest ground-based telescopic observatory is located at :

- (a) Colombia (b) India
(c) Nepal (d) Switzerland

I.A.S. (Pre) 2002

Ans. (*)

Presently, 'The University of Tokyo Atacama Observatory' is the highest telescope observatory in the world which is located on the summit of Cerro Chajnantor at an altitude of 5,640 m within a lava dome in the Atacama Desert of northern Chile. But in the question period, Indian Astronomical Observatory was considered as the world's highest telescope observatory which is located in Hanley village in Ladakh at an altitude of 4,500 m.

58. Consider the following statements about recently inaugurated Asia's largest and world's highest imaging Cherenkov Observatory, at Hanle, Ladakh :

- I. It was inaugurated by Ajit Kumar Mohanty, Chairman, Atomic Energy Commission.
- II. It is located at an altitude of 5100 metres.
- III. The telescope is built by Bhabha Atomic Research Centre with the support of Electronic Corporation of India and other industry partners.

Choose the correct answer from the options given below:

- (a) Only I and II are correct.
(b) I, II and III are correct.
(c) Only I and III are correct.
(d) Only II and III are correct.

Rajasthan P.C.S. (Pre) 2024

Ans. (c)

Dr. Ajit Kumar Mohanty, Secretary DAE & Chairman of the Atomic Energy Commission, inaugurated the Major Atmospheric Cherenkov Experiment (MACE) Observatory at Hanle, Ladakh, on 4th October, 2024. MACE is the largest imaging Cherenkov telescope in Asia. Located at an altitude of $\sim 4,270$ m above sea level, it is also the highest of its kind in the world. The telescope is indigenously built by Bhabha Atomic Research Centre (BARC) with support from Electronic Corporation of India Ltd. (ECIL) and other Indian industry partners. Hence, statement I and III are correct while statement II is incorrect.

59. What is the full name of ISRO?

- (a) International Space Research Organisation
(b) Indian Society Research Organisation
(c) Iranian Space Research Organisation
(d) Indian Space Research Organisation

M.P. P.C.S. (Pre) 2021

Ans. (d)

Indian Space Research Organisation (ISRO) is the space agency of India. The organisation is involved in science, engineering and technology to harvest the benefits of outer space for India and the mankind. ISRO is a major constituent of the Department of Space (DOS), Government of India. ISRO was previously the Indian National Committee for Space Research (INCOSPAR), set up by the Government of India in 1962, as envisioned by Dr. Vikram A. Sarabhai. ISRO was formed on August 15, 1969 and superseded INCOSPAR with an expanded role to harness space technology.

60. Who amongst the following was the Chairman of ISRO when INSAT-3B was launched ?

- (a) Anil Kakodkar (b) Abdul Kalam
(c) K. Kasturirangan (d) U.R. Rao

R.A.S./R.T.S. (Pre) 2001

Ans. (c)

Dr. K. Kasturirangan headed the ISRO from 1994 to 2003. During his tenure, India launched INSAT-3B satellite with the help of Ariane 5G rocket from the Kourou launching station of French Guiana on 21 March, 2000.

61. Chairman of Indian Space and Research Organization is :

- (a) Krishna Swamy Kasturirangan
(b) A.P.J. Abdul Kalam
(c) G. Madhavan Nair
(d) K.M. Bhandari

R.A.S./R.T.S.(Pre) 2008

Ans. (*)

During the period, when the question has been asked, Dr. G. Madhavan Nair was the Chairman of ISRO. At present Dr. V. Narayanan is the Chairman of ISRO. He assumed this office on 13th January, 2025.

62. Who is the new Chairman of 'ISRO' from the following?

- (a) R. Hatten (b) A.S. Kiran Kumar
(c) S. Sivan (d) P. Kulikrishnan
(e) None of the above

Chattisgarh P.C.S. (Pre) 2018

Ans. (e)

A.S. Kiran Kumar was the Chairman of ISRO from January, 2015 to January, 2018. The 10th ISRO Chairman was Dr.S. Somanath whose tenure was from January, 2022 to January, 2025. The present (11th) Chairman of ISRO is Dr. V. Narayanan who assumed the charge on 13th January, 2025.
The first Chairman of the ISRO was Dr. Vikram A. Sarabhai.

63. Who among the following was appointed as the tenth Chairman of the Indian Space Research Organisation (ISRO) in January 2022?

- (a) P.L.N. Raju (b) V. Hari Prasad
(c) C. Jagannathan (d) S. Somanath

U.P. R.O./A.R.O. (Mains) 2021

Ans. (d)

See the explanation of above question.

64. Who is the current Chairman of the Indian Space Research Organisation (ISRO)?

- (a) G. Madhavan Nair (b) U. R. Rao
(c) S. Somanath (d) K. Sivan

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (*)

The current Chairman of the Indian Space Research Organisation (ISRO) is Dr. V. Narayanan. He assumed the charge of Secretary, Department of Space, Chairman, Space Commission and Chairman, ISRO on 13th January, 2025. He succeeded Dr. S. Somanath, who hold these positions from January, 2022 to January, 2025.

65. Assertion (A) : India's telecommunication has been linked to a satellite.

Reason (R) : India is preparing for starwar.

- (a) Both (A) and (R) are true, but (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 1990

Ans. (c)

INSAT is a series of multipurpose geostationary satellites launched by ISRO to fulfill the needs in the field of telecommunications, broadcasting, meteorology etc. Thus Assertion (A) is true, but Reason (R) is false because India is not preparing for starwar.

66. Indian Satellite series which is being used for telecommunication system is

- (a) Rohini (b) I.R.S.
(c) CARTOSAT (d) INSAT

Uttarakhand P.C.S. (Pre) 2016

Ans. (d)

See the explanation of above question.

67. Indian Satellite Series 'INSAT' is an example of :

- (a) Resource Satellite
(b) Military Satellite
(c) Polar orbiting Satellite
(d) Meteorological Satellite

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

See the explanation of above question.

68. Which of the following frequency bands is not used in INSAT/GSAT satellite communication?

- (a) Ka (b) C
(c) MF (d) Ku

R.A.S./ R.T.S. (Pre) 2021

Ans. (c)

The major frequency bands used for satellite communication in India are S-band, C-band, Upper Extended C band, Ku band and Ka band. The INSAT/GSAT satellites carry transponders in these bands.

69. Television transmission is an example of which of the following?

- (a) Simplex communication
(b) Half-duplex communication
(c) Full-duplex communication
(d) None of the above

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

Television transmission is an example of simplex communication. Simplex is a one-way only communication in which broadcasted information travel in one direction only.

70. India's first remote sensing satellite (I.R.S.-IA) was launched from :

- (a) Baikanour (b) Cape Kennedy
(c) French Guana (d) Sri Harikota

U.P.P.C.S. (Pre) 1994

Ans. (a)

The first Indian Remote-sensing satellite IRS-IA was launched on March 17, 1988 from Soviet Union Baikanour Cosmodrome.

71. In which of the following activities are Indian Remote Sensing (IRS) satellites used?

1. Assessment of crop productivity
2. Locating groundwater resources
3. Mineral exploration
4. Telecommunications
5. Traffic studies

Select the correct answer using the code given below.

- (a) 1, 2 and 3 only
(b) 4 and 5 only
(c) 1 and 2 only
(d) 1, 2, 3, 4 and 5

I.A.S. (Pre) 2015

Ans. (a)

The remote sensing applications using IRS have proliferated into almost every aspect of national development. The data is now used regularly for agricultural crop acreage and yield estimation, drought monitoring and assessment, flood mapping, land use and land cover mapping, wasteland management, water resources management, ocean/marine resources survey and management, mineral prospecting, forest resources survey and management etc. Remote sensing satellites are not used for telecommunications or traffic studies.

72. For the measurement/estimation of which of the following are satellite images/remote sensing data used?

1. Chlorophyll content in the vegetation of a specific location
2. Greenhouse gas emissions from rice paddies of a specific location
3. Land surface temperatures of a specific location

Select the correct answer using the code given below.

- (a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2019

Ans. (d)

For the measurement/estimation of chlorophyll content in the vegetation of a specific location (through hyper-spectral and multi-spectral imaging), greenhouse gas emissions from rice paddies of a specific location (through short wave infrared, thermal infrared and Merlin Lidar techniques) and land surface temperatures of a specific location (through thermal infrared etc.), satellite images/remote sensing data are used.

73. INSAT-3C was launched from-

- (a) Hassan (b) Kourou
(c) Sriharikota (d) Thumba

U.P. Lower Sub. (Pre) 2002

Ans. (b)

INSAT-3C was launched by the Ariane-4 launch vehicle of Arianespace from Kourou, French Guiana in South America. Communication satellite GSAT-30 was launched from Kourou on 17 January, 2020 while communication satellite CMS-01 was launched by PSLV-C50 from Sriharikota on 17 December, 2020. CMS-02 (GSAT-24) was launched on 22 June, 2022 by Ariane 5 rocket from Kourou, French Guiana. Communication satellite GSAT-N2 was successfully launched on 19 November, 2024, aboard a Falcon-9 rocket from Cape Canaveral Space Force Station, Florida, USA.

74. Which Satellite was successfully launched by ISRO on December 22, 2005 ?

- (a) INSAT - 2A (b) INSAT - 2B
(c) INSAT - 4A (d) INSAT - 4B

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (c)

INSAT-4A, the first one in INSAT-4 satellites series was successfully launched by ISRO on December 22, 2005 from Kourou, French Guiana by Ariane-5V169 launch vehicle. The satellite weighed 3081 kg.

75. How many satellites were launched successfully by PSLV-C37 in a single flight on 15 February, 2017?

- (a) 80 (b) 104
(c) 72 (d) 98

R.A.S./R.T.S. (Pre) 2018

Ans. (b)

In its 39th flight (PSLV-C37), ISRO's Polar Satellite Launch Vehicle successfully launched the 714 kg Cartosat-2 series satellite along with 103 co-passengers satellites on 15 February, 2017 from Satish Dhawan Space Center SHAR, Sriharikota. The total weight of all the 104 satellites carried on board PSLV-C37 was 1378 kg. This was a record for the highest number of satellites sent to space on a single launch, until 24 January 2021, when SpaceX launched the Transporter-1 mission on a Falcon 9 rocket carrying 143 satellites into orbit.

76. In 2017, which Satellite Launch Vehicle of ISRO successfully launched 104 satellites in a single flight?

- (a) INSAT-3DR (b) PSLV-C55
(c) GSLV-F12 (d) PSLV-C37

M.P. P.C.S. (Pre) 2024

Ans. (d)

See the explanation of above question.

77. What is the name of last PSLV mission of ISRO which was launched in 2022?

- (a) PSLV-C54 (b) PSLV-C53
(c) PSLV-C52 (d) PSLV-C51

Chhattisgarh P.C.S. (Pre) 2022

Ans. (a)

The last PSLV mission by ISRO in 2022 was PSLV-C54/ EOS-06, which was launched from Satish Dhawan Space Centre, SHAR, Sriharikota on 26 November, 2022. After that, following PSLV missions are launched by the ISRO till April, 2025 :

22 April, 2023	–	PSLV-C55/TeLEOS-2 Mission
30 July, 2023	–	PSLV-C56/DS-SAR Mission
2 September, 2023	–	PSLV-C57/Aditya-L1 Mission
1 January, 2024	–	PSLV-C58/XPoSat Mission
5 December, 2024	–	PSLV-C59/Proba-3 Mission
30 December, 2024	–	PSLV-C60/SpaDeX (A&B) Mission

78. Satellites TeLEOS-2 and Lumelite-4 were launched in April 23 from Sriharikota by the launcher :

- (a) SSLV - D2 (b) LVM3 - M4
(c) PSLV - C55 (d) GSLV - F12

Rajasthan P.C.S. (Pre) 2023

Ans. (c)

The PSLV-C55 was the 57th mission of Indian Space Research Organisation's Polar Satellite Launch Vehicle (PSLV). PSLV - C55/TeLEOS-2 was launched successfully on 22 April, 2023 from Satish Dhawan Space Centre (SDSC), SHAR, Sriharikota, Andhra Pradesh. This was a dedicated commercial mission through NewSpace India Limited (NSIL) with TeLEOS-2 as primary satellite and Lumelite-4 as a co-passenger satellite. The satellites weigh about 741 kg and 16 kg, respectively. Both satellites belong to Singapore.

79. Consider the following statements about recently launched XPoSat (X-ray Polarimeter Satellite) by ISRO.

- (1) It is India's first dedicated polarimetry mission to study various dynamics of bright astronomical X-ray sources in extreme conditions.
(2) The spacecraft will carry two scientific payloads in a low earth orbit.

Which of the statement/s given above is/are true?

- (a) Only (1) (b) Only (2)
(c) Both (1) and (2) (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (c)

ISRO's PSLV-C58 has launched XPoSat Satellite on 1 January, 2024. XPoSat (X-ray Polarimeter Satellite) is India's first dedicated polarimetry mission to study various dynamics of bright astronomical X-ray sources in extreme conditions. The spacecraft carried two scientific payloads in a low earth orbit. The primary payload POLIX (Polarimeter Instrument in X-rays) will measure the polarimetry parameters (degree and angle of polarization) in medium X-ray energy range of 8–30 keV photons of astronomical origin. The XSPECT (X-ray Spectroscopy and Timing) payload will give spectroscopic information in the energy range of 0.8 – 1.5 keV.

80. Which is the correct chronological order of the following space programmes?

- A. APPLE B. IGMD
C. STEP D. SITE

Code :

- (a) B, C, D, A (b) D, C, A, B
(c) C, A, D, B (d) A, C, D, B

U.P.P.C.S. (Pre) 1998

Ans. (b)

SITE – Satellite Instructional Television Experiment project launched in 1975, jointly designed by NASA and ISRO.

STEP – Satellite Telecommunication Experiment Project (launched in 1977).

APPLE – Ariane Passenger Payload Experiment was launched on June 19, 1981 by Ariane launch vehicle from Kourou in French Guiana.

IGMD – Integrated Guided Missile Development Program which was launched by DRDO in 1983.

81. Apple was launched from –

- (a) Kourou (French Guyana)
(b) Sriharikota (India)
(c) Baikanoor (Russia)
(d) Cape Canaveral (U.S.A.)

Uttarakhand P.C.S. (Pre) 2007

Ans. (a)

Apple Satellite was launched on June 19, 1981 by Ariane launch vehicle from Kourou in French Guiana. It was India's First experimental geostationary communication satellite.

82. The first Indian Communication Satellite APPLE was launched on :

- (a) Feb 27, 1982 (b) September 18, 1981
(c) June 19, 1981 (d) December 25, 1981

Chhattisgarh P.C.S. (Pre) 2008

Ans. (c)

See the explanation of above question.

83. When INSAT I-D of India was launched ?

- (a) June 10th, 1990 (b) June 12th, 1990
(c) June 13th, 1990 (d) June 15th, 1990

U.P.P.C.S. (Pre) 1994

Ans. (b)

INSAT - 1D was launched on June 12th, 1990. INSAT-1D was a multifunctional satellite that provides telephone, television, and weather observation service for India.

84. When was 'INSAT-2A' launched?

- (a) February 4th, 1975 (b) May 18th, 1974
(c) July 10th, 1992 (d) April 19th, 1975

44th B.P.S.C. (Pre) 2000

Ans. (c)

INSAT-2A was India's first indigenously built satellite. This satellite was launched on July 10th, 1992 from Kourou, French Guiana.

85. Heaviest Indian Satellite, INSAT-4C, which failed to go into orbit recently weighted ?

- (a) 1568 kg (b) 2168 kg
(c) 2568 kg (d) 3068 kg

U.P.P.C.S. (Mains) 2005

Ans. (b)

On July 10, 2006 the Geosynchronous Satellite Launch Vehicle (GSLV) carrying the INSAT-4C communication satellite, veered off course and exploded 60 seconds after lift-off. It had a weight of 2168 kg and developed by ISRO with a cost of Rs. 350 crore. At present, weighing about 5854 kg, GSAT-11 is the heaviest satellite built by ISRO. It was launched on 5 December, 2018 from Kourou launch base, French Guiana by Ariane-5 VA-246.

86. India's first dedicated navigation satellite IRNSS-1A was launched from :

- (a) Sriharikota (b) Ahmedabad
(c) Thiruvananthapuram (d) Bengaluru

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

ISRO's Polar Satellite Launch Vehicle PSLV-C22 successfully launched IRNSS-1A, the first satellite in the Indian Regional Navigation Satellite System (IRNSS) on July 1st, 2013 from Satish Dhawan Space Centre, Sriharikota.

87. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements :

1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

- (a) 1 only (b) 1 and 2 only

- (c) 2 and 3 only (d) None

I.A.S. (Pre) 2018

Ans. (a)

The Indian Regional Navigation Satellite System (IRNSS) named by P.M. Modi as NavIC (Navigation with Indian Constellation), is a regional satellite navigation system that provides accurate real time positioning and timing services covering India and region extending to 1500 km around. In IRNSS, out of seven satellites of constellation, three satellites were deployed in geostationary orbit and four satellites were placed in geosynchronous orbit. There is no proposed plan to extend it throughout the world. Thus, only statement 1 is correct.

88. Full form of NavIC is :

- (a) Navigation without International Control
(b) Navigation with Indian Constellation
(c) Navigation with Indian Coastal
(d) Navigation with Indian Cooperation

Chhattisgarh P.C.S. (Pre) 2021

Ans. (b)

See the explanation of above question.

89. Which one of the following countries has its own Satellite Navigation System?

- (a) Australia (b) Canada
(c) Israel (d) Japan

I.A.S. (Pre) 2023

Ans. (d)

Currently, there are four global Satellite Navigation Systems viz., GPS from USA, GLONASS from Russia, Galileo from European Union and BeiDou from China. In addition, there are two regional Satellite Navigation Systems viz., NavIC from India and QZSS (Michibiki) from Japan.

90. What is the Galileo Project which has been in news recently?

- (a) An inter-country programme of missile shield developed by the United States of America
(b) A project developed by India with assistance from Canada
(c) An environmental protection project being developed by Japan
(d) A multi-satellite navigation project being developed by the European Union

I.A.S. (Pre) 2007

Ans. (d)

Galileo is Europe's own global multi-satellite navigation system (which went live in 2016), providing a highly accurate and guaranteed global positioning service under civilian control. It is inter-operable with GPS, GLONASS and other global satellite navigation systems. It is created by the European Union through European Space Agency (ESA).

91. GPS stands for which one of the following ?

- (a) Greenwich Polar Satellite
(b) Global Police Surveillance
(c) Global Positioning System
(d) General Pacific Survey

U.P.P.C.S. (Mains) 2015

Ans. (c)

GPS means Global Positioning System, is a space-based navigation system. It was developed by the U.S. Department of defence. It became fully operational from 27th April, 1995.

92. In which of the following areas can GPS technology be used?

1. Mobile phone operations
2. Banking operations
3. Controlling the power grids

Select the correct answer using the codes given below:

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (d)

The Global Positioning System (GPS) is a satellite-based radio navigation system that provides geo-location and time information to a GPS receiver anywhere on or near the Earth. GPS technology is now in everything from cellphones and wristwatches to bulldozers, shipping containers and ATM's. Major communication networks, banking systems, financial market and power grids depend heavily on GPS for precise time synchronization.

93. India developed the Satellite launch vehicles over a period of few decades. Which is the correct chronological order of development of the following vehicles A, B, C and D ?

Select correct answer from Code below :

- A. PSLV B. ASLV
C. SLV D. GSLV

Code :

- (a) A, B, C, D (b) B, C, D, A

(c) D, C, A, B

(d) C, B, A, D

U.P.P.C.S. (Mains) 2004

Ans. (d)

The correct chronological order of development of above satellite launch vehicles are –
SLV – August 10, 1979
ASLV – March 24, 1987
PSLV – September 20, 1993
GSLV – April 18, 2001
Therefore the correct option is (d).

94. Match the following pairs :

- | Launch Vehicle | Satellite |
|---|------------------|
| (i) Satellite Launch Vehicle-3 (SLV-3) | A. Chandrayaan-1 |
| (ii) Augmented Satellite Launch Vehicle (ASLV) | B. Rohini |
| (iii) Polar Satellite Launch Vehicle (PSLV) | C. SROSS-C |
| (iv) Geosynchronous Satellite Launch Vehicle (GSLV) | D. EDUSAT |

Which of the following codes are correctly matched?

Code :

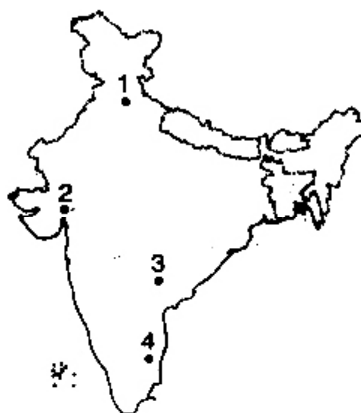
- | | (i) | (ii) | (iii) | (iv) |
|-----|-----|------|-------|------|
| (a) | A | B | C | D |
| (b) | B | C | D | A |
| (c) | B | C | A | D |
| (d) | C | A | B | D |

R.A.S./R.T.S. (Pre) 2016

Ans. (c)

SLV-3 was successfully launched on July 18, 1980 when Rohini satellite, RS-1 was placed in the orbit, thereby making India the sixth member of an exclusive club of space-faring nations. The Augmented Satellite Launch Vehicle (ASLV) Programme was designed to augment the payload capacity to 150 kg thrice that of SLV-3 for low earth orbits. The third developmental flight ASLV-D3 was successfully launched on May 20, 1992, when SROSS-C (106 kg) was put into the orbit of 225×430 km. PSLV is one of the world's most reliable launch vehicles. It has been in service for twenty years and has launched various satellites for historic missions like Chandrayaan-1, Mars Orbiter Mission, Indian Regional Navigation Satellite System (IRNSS), etc. The Geosynchronous Satellite Launch Vehicle (GSLV) was primarily developed to launch INSAT/GSAT class of satellites into orbit. GSLV is being used for launching GSAT series of the satellite. GSLV has launched various communication satellites among which EDUSAT is notable, being India's first dedicated educational satellite. GSLV-MkIII (LVM3) has launched Chandrayaan-2 and Chandrayaan-3.

95. In the given picture/map 1, 2, 3 and 4 has marked to show the location of space organization units. Match these units with the list given below and select the correct answer using the codes given below :



List :

- (A) I.S.R.O.
(C) N.R.S.A.

- (B) I.I.R.S.
(D) S.A.C.

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 1 | 2 | 3 |
| (b) | 1 | 4 | 3 | 2 |
| (c) | 1 | 4 | 2 | 3 |
| (d) | 4 | 1 | 3 | 2 |

I.A.S. (Pre) 2001

Ans. (d)

IIRS–Indian Institute of Remote Sensing, Dehradun (Uttarakhand)
SAC– Space Applications Centre, Ahmedabad (Gujarat)
NRSA–National Remote Sensing Agency (Now NRSC : National Remote Sensing Centre), Hyderabad (Telangana)
ISRO–Indian Space Research Organisation, Bengaluru (Karnataka)

96. ISRO's Master Control Facility is at Hasan, which is located in :

- | | |
|--------------------|---------------|
| (a) Andhra Pradesh | (b) Gujarat |
| (c) Orissa | (d) Karnataka |

U.P.P.S.C. (GIC) 2010

Ans. (d)

The Master Control Facility (MCF) is a facility set up by the ISRO in Karnataka. It was established in 1982 for monitoring and controlling geostationary and geosynchronous satellites launched by ISRO. Another MCF was established in Bhopal in 2005.

97. Match List- I with List- II and select the correct answer from the code given below the Lists :

List- I (Organisations)	List- II (Location)
A. ISRO	1. Thiruvananthapuram
B. IUCA	2. Pune
C. IUAC	3. Bengaluru
D. VSSC	4. New Delhi

Code :

	A	B	C	D
(a)	3	2	4	1
(b)	1	2	3	4
(c)	2	4	1	3
(d)	3	1	2	4

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (a)

Vikram Sarabhai Space Centre is located in Thiruvananthapuram, Kerala. ISRO's headquarter is located in Bengaluru, Karnataka. IUAC (Inter-University Accelerator Centre) is established in New Delhi. IUCA (Inter-University Centre for Astronomy and Astrophysics) is established in Pune, Maharashtra.

98. Which of the following is not correctly matched?

- (a) Space Applications Centre – Ahmedabad
- (b) National Remote Sensing – Hyderabad Centre
- (c) Indian Space Research – Sriharikota Organisation Satellite Centre
- (d) Vikram Sarabhai Space – Thiruvananthapuram Centre

U.P.U.D.A./L.D.A (Spl.) (Mains) 2010

Ans. (c)

The ISRO Satellite Centre (Renamed as 'UR Rao Satellite Centre : URSC' in 2018) was established originally in 1972 as Indian Scientific Satellite Project (ISSP) at Bengaluru, Karnataka. It became ISRO Satellite Centre in 1976. Other pairs are correctly matched.

99. Which one of the following pairs are correctly matched:

- (a) Vikram Sarabhai Space Centre – Sriharikota
- (b) ISRO Satellite Centre – Thumba
- (c) SHAR Centre – Bangalore
- (d) Space Applications Centre – Ahmedabad

U.P.P.C.S.(Pre) 1998

Ans. (d)

Space Applications Centre is located in Ahmedabad. While ISRO Satellite centre (now URSC) is located in Bengaluru, Vikram Sarabhai Space Centre is in Thiruvananthapuram and SHAR centre is located in Sriharikota.

100. Consider the following pairs :

ISRO Centres	Location
A. Vikram Sarabhai Space Centre (VSSC)	– Trivandrum (Kerala)
B. UR Rao Satellite Centre (URSC)	– Bengaluru (Karnataka)
C. National Remote Sensing Centre (NRSC)	– Dehradun (Uttarakhand)
D. Indian Institute of Remote Sensing (IIRS)	– Hyderabad (Telangana)

Which of the above pairs are correctly matched?

- (a) A & B only
- (b) A, B & C only
- (c) B, C & D only
- (d) A, B, C & D

R.A.S./R.T.S. (Pre) 2018

Ans. (a)

The Vikram Sarabhai Space Centre is situated at Trivandrum (Kerala) and UR Rao Satellite Centre is situated at Bengaluru (Karnataka). While the National Remote Sensing Centre is situated at Hyderabad (Telangana) and Indian Institute of Remote Sensing is situated at Dehradun (Uttarakhand). So, only A and B pairs are correctly matched.

101. The Indian Institute of Remote Sensing (IIRS) is situated at :

- (a) Dehradun
- (b) New Delhi
- (c) Lucknow
- (d) Nagpur

Uttarakhand P.C.S. (Pre) 2012

Chhattisgarh P.C.S. (Pre) 2018

U.P.U.D.A./L.D.A. (Pre) 2006

Ans. (a)

See the explanation of above question.

102. The Vikram Sarabhai Space Centre at Trivandrum is mainly concerned with –

- 1. Research in propellants for launching rockets.
- 2. To launched sounding rockets to investigate problems of meteorology.

3. Design and fabrication of spacecraft.

4. Development of satellite launch.

Select the answer from the code given below :

Code :

- (a) 1 and 4 only (b) 1 and 2 only
(c) 1 and 3 only (d) 1 only

U.P.P.C.S. (Mains) 2004

Ans. (a)

The Vikram Sarabhai Space Centre at Trivandrum (Thiruvananthapuram) is mainly concerned with the research in propellant for launching rockets and development of satellite launch. The space technology research in India began with the establishment of Indian Space Research Organisation in 1969. Dr. Vikram Sarabhai is credited for giving new directions in the field of Indian space program. In fact, he is widely regarded as the father of Indian space program.

103. With reference to 'PraVaHa' software developed by ISRO, which of the following statements is/are correct?

- 1. It is a path-breaking feat by Vikram Sarabhai Space Centre.**
2. Initial aerodynamic design of space vehicles requires evaluation of large number of configurations.

Select the correct answer using the codes given below:

- (a) Only 2 (b) Neither 1 nor 2
(c) Only 1 (d) Both 1 and 2

U.P.P.C.S. (Pre) 2024

Ans. (d)

PraVaHa is a Computational Fluid Dynamics (CFD) software developed by ISRO (Indian Space Research Organisation) for aerodynamic design and analysis particularly for launch vehicles and re-entry vehicles. The CFD software PraVaHa (Parallel RANS Solver for Aerospace Vehicle Aero-thermodynamic Analysis) is developed in ISRO's Vikram Sarabhai Space Centre (VSSC), Thiruvananthapuram (the lead centre of ISRO responsible for the design and development of launch vehicle technology), to simulate external and internal flows on launch vehicles, winged and nonwinged re-entry vehicles. Initial aerodynamic design studies for launch vehicles demand evaluation of a large number of configurations. Any aerospace vehicle while moving through the Earth's atmosphere during 'launch' or 're-entry' is subjected to severe aerodynamic and aerothermal loads in terms of external pressure and heat flux. PraVaHa has been used extensively in the Gaganyaan program for aerodynamic analysis of human-rated launch vehicles, viz. HLVM3, Crew Escape System [CES], and Crew Module [CM]. It is designed to make use of CPU as well as GPU architecture of available and upcoming supercomputing facilities. Hence, both statements are correct.

104. Thumba known as rocket launching station is situated in :

- (a) Uttar Pradesh (b) Bihar
(c) Jharkhand (d) Kerala

U.P.P.C.S. (Mains) 2004

Ans. (d)

Thumba Equatorial Rocket Launch Station (TERLS) was established in 1962 at Thiruvananthapuram, Kerala.

105. Sriharikota Island lies near :

- (a) Chika lake (b) Pulicat lake
(c) Mahanadi (d) Godavari Mouth

U.P.P.C.S. (Mains) 2003

Ans. (b)

Sriharikota is a barrier island in the Andhra Pradesh. It is most famous as the home of India's only satellite launch centre. Sriharikota divides the Pulicat Lake from the Bay of Bengal.

106. In which of the following States lies Sriharikota, the spaceport of India ?

- (a) Maharashtra (b) Andhra Pradesh
(c) Tamil Nadu (d) Kerala

Uttarakhand P.C.S. (Pre) 2002

M.P.P.C.S. (Pre) 1997

Ans. (b)

See the explanation of above question.

107. Match List-I with List-II and select the correct answer from the code given below the lists.

List-I (Place)	List-II (Function)
A. Kakrapar	1. Atom bomb blast site
B. Pokhran	2. Rocket launching centre
C. Thumba	3. Nuclear power plant
D. Sri Harikota	4. Satellite launching station

Code :

A	B	C	D
(a) 3	2	1	4
(b) 3	2	4	1
(c) 3	1	2	4
(d) 4	3	2	1

U.P. R.O./A.R.O. (Mains) 2016

Ans. (c)

The correctly matched lists are as follows :

List-I (Place)	List-II (Function)
Kakrapar	– Nuclear power plant
Pokhran	– Atom bomb blast site
Thumba	– Rocket launching centre
Sri Harikota	– Satellite launching station

108. In February, 2004 a blast occurred in India's Space Centre at :

- (a) Ahmedabad (b) Bangalore
(c) Sriharikota (d) Thumba

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

In February, 2004 a blast occurred at a plant making solid propellant at the Satish Dhawan Space Centre, Sriharikota.

109. Consider the following Assertion (A) and Reason (R) and choose your answer from the given code :

Assertion (A) : The upper surface of the wings of an aeroplane is made convex and the lower surface is made concave.

Reason (R) : The air current at the top has less velocity and thus less pressure at the bottom than at the top.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is false and (R) is true.
(d) (A) is true and (R) is false.

U.P.P.C.S. (Mains) 2013

Ans. (d)

The upper surface of the wings of an aeroplane is made convex and built concave at lower surface. Such design of the wings makes the air velocity faster at the upper surface than the lower surface, which creates low pressure at the upper side. This pressure difference provides lift to the wings of the plane.

110. The 'Black Box' of an aeroplane is of :

- (a) White colour (b) Red colour
(c) Black colour (d) Orange colour

Uttarakhand P.C.S. (Pre) 2010

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Spl.) (Mains) 2004

U.P.P.C.S. (Pre) 1996

Ans. (d)

The black boxes are actually painted with a heat-resistant bright orange colour in order to make them easier to find in wreckage resulting from the crash. It is a flight data recorder and the cockpit voice recorder of an aircraft. It is an extremely important tool for investigators in the event of an air crash.

111. The Moon is :

- (a) A Star (b) A Planet

(c) A Satellite

(d) A Meteorite

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

The Moon is Earth's only natural satellite. It is fifth largest satellite in our solar system.

112. On which date did Chandrayaan enter the lunar orbit?

- (a) 3 November (b) 6 November
(c) 8 November (d) None of these

M.P.P.C.S. (Pre) 2010

Ans. (c)

Chandrayaan-1, India's first mission to the Moon was launched successfully on October 22, 2008 from Satish Dhawan Space Centre, Sriharikota (Andhra Pradesh) and the vehicle was successfully inserted into lunar orbit on November 8, 2008. This mission was operational till August 29, 2009. Chandrayaan-2 mission was launched on 22 July, 2019 and it reached the lunar orbit on 20 August, 2019. While Chandrayaan-3 mission was launched on 14 July, 2023 and it entered lunar orbit on 5 August, 2023.

113. Name of the spacecraft for India's mission Moon is :

- (a) Chandrayaan-I (b) Chandrayaan-II
(c) Antariksha (d) Change-I

M.P.P.C.S. (Pre) 2008

Ans. (a) & (b)

See the explanation of above question.

114. CHANDRAYAAN-I was launched from :

- (a) Orissa (b) Tamil Nadu
(c) Karnataka (d) Andhra Pradesh

R.A.S./R.T.S. (Pre) 2008

Ans. (d)

See the explanation of above question.

115. The Project Director of Chandrayaan-2 is :

- (a) C. Rangarajan (b) A.P.J. Kalam
(c) P. Chidambaram (d) Dr. M. Annadurai

U.P.P.S.C. (GIC) 2010

Ans. (d)

Chandrayaan-2 was an ISRO mission aimed at placing an orbiter around the moon and sending a lander and a rover to its surface. Chandrayaan-2 project was initially headed by Dr. M. Annadurai, also known for leading the Chandrayaan-1 project. Finally, Chandrayaan-2 mission has been headed by two women scientists of ISRO—Muthayya Vanitha as Project Director and Ritu Karidhal as Mission Director. Chandrayaan-2 mission was launched from the Satish Dhawan Space Centre on 22 July, 2019 to the moon by GSLV-MkIII (LVM3) rocket.

116. What is not true about Chandrayaan-2 Mission?

- (a) It was approved by Prime Minister Manmohan Singh
- (b) Rover sent in this mission was named Pragyan
- (c) In this mission Lunar North Pole was targetted for exploration
- (d) Vikram Lander was used in this mission

Chhattisgarh P.C.S. (Pre) 2021

Ans. (c)

Chandrayaan-2 Mission was originally approved in 2008 in a meeting of the Union Cabinet, chaired by the then Prime Minister Manmohan Singh. Rover sent in this mission was named Pragyan while the lander used in this mission was named Vikram. The lander and rover were scheduled to land on the near side of the Moon, in the south polar region. Hence, statement of option (c) is incorrect.

117. In July 2019, the spacecraft Chandrayaan-2 was launched from which Indian State?

- (a) Gujarat
- (b) Andhra Pradesh
- (c) Karnataka
- (d) Maharashtra
- (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Chandrayaan-2 is the second lunar exploration mission developed by the Indian Space Research Organisation (ISRO). This spacecraft was launched on its mission to the moon from the Satish Dhawan Space Centre, Sriharikota (Andhra Pradesh) on 22 July, 2019 by GSLV-MK III- M1. The spacecraft reached the moon's orbit on 20 August, 2019. Its lander Vikram with lunar rover Pragyan were scheduled to land on the moon on 7 September, 2019 but lander deviated from its intended trajectory starting at 2.1 km altitude and had lost communication. A successful soft landing would have made India fourth country after Russia, USA and China to do so. However, lunar orbiter of Chandrayaan-2 is working well and has been in operation for over 5 years and 8 months (as of 30 April, 2025).

118. Which Geosynchronous Satellite Launch Vehicle was used by ISRO to launch Chandrayaan-2 spacecraft?

- (a) GSLV-MkIII-M1
- (b) GSLV-MkII-M2
- (c) GSLV-MkIV-M8
- (d) GSLV-MkV-M4

U.P.P.C.S. (Pre) 2019

Ans. (a)

Geosynchronous Satellite Launch Vehicle GSLV-MkIII (LVM3)-M1 was used by ISRO to launch Chandrayaan-2 spacecraft. It is a three-stage heavy lift launch vehicle developed by ISRO. The vehicle has two solid strap-ons, a core liquid booster and a cryogenic upper stage. GSLV-MkIII (LVM3) is designed to carry 4 ton class of satellites into Geosynchronous Transfer Orbit (GTO) or about 10 tons to Low Earth Orbit (LEO), which is about twice the capability of the GSLV-MkII.

119. Which of the following statements is/are true about Chandrayaan-3 of ISRO?

(1) Chandrayaan-3 is a follow-on mission to Chandrayaan-2 to demonstrate end-to-end capability in safe landing and roving on the lunar surface.

(2) It was launched by Launch vehicle LVM3.

- (a) Only (1)
- (b) Only (2)
- (c) Both (1) and (2)
- (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (c)

Chandrayaan-3 is the third mission in the Chandrayaan programme, a series of lunar-exploration missions developed by the Indian Space Research Organisation (ISRO). Chandrayaan-3 is a follow-on mission to Chandrayaan-2 to demonstrate end-to-end capability in safe landing and roving on the lunar surface. It consists of Lander and Rover configuration. The mission consists of a Vikram lunar lander and a Pragyan lunar rover similar to those launched aboard Chandrayaan-2 in 2019, as well as a propulsion module that carried the spacecraft from Earth orbit to lunar orbit. It was launched by LVM3-M4 from SDSC, SHAR, Sriharikota on 14 July, 2023. The spacecraft entered lunar orbit on 5 August, 2023 and became the 1st lander to touch down near the lunar south pole on 23 August, 2023 making India the fourth country to successfully land on the Moon. Hence, both statements are correct.

120. Which one of the following payloads was used in Chandrayaan-3 Mission to determine the elemental composition (Mg, Al, Si, K, Ca, Ti, Fe) of lunar soil and rocks around the lunar landing site?

- (a) ILSA
- (b) LRA
- (c) APXS
- (d) None of the above

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

Chandrayaan-3 rover 'Pragyan' has two payloads : Alpha Particle X-ray Spectrometer (APXS) and Laser Induced Breakdown Spectroscope (LIBS). Among them, APXS was used to determine the elemental composition (Mg, Al, Si, K,

Ca, Ti, Fe) of lunar soil and rocks around the lunar landing site, while LIBS was used for qualitative and quantitative elemental analysis and to derive the chemical composition and infer mineralogical composition to further our understanding of lunar surface.

121. With reference to 'Lunar Polar Exploration Mission' (LUPEX), which of the following statements is/are correct?

1. This is an India-Japan partnership for their joint moon mission which shall be launched in the future.
2. JAXA is responsible for developing the lander for the mission, while ISRO is handling the rover and launch vehicle.

Select the correct answer using the codes given below:

- (a) Only 2 (b) Neither 1 nor 2
(c) Only 1 (d) Both 1 and 2

U.P.P.C.S. (Pre) 2024

Ans. (c)

The Lunar Polar Exploration (LUPEX) Mission, a collaborative effort between India's Indian Space Research Organisation (ISRO) and Japan's Japan Aerospace Exploration Agency (JAXA), is an innovative mission focused on investigating the Moon's south pole. This mission represents a notable advancement in lunar research and has significant implications for future space endeavours, including the potential for a permanent human presence on the Moon. Scheduled for launch in the late 2020s, LUPEX will enhance our understanding of critical resources on the Moon's surface, particularly concentrating on water ice deposits. In this partnership, JAXA is tasked with creating a sophisticated rover that can traverse and perform research on the Moon's surface, while ISRO is set to provide a reliable lander for deploying the rover. Furthermore, the mission will include instruments from NASA and the European Space Agency (ESA), allowing for a wide variety of scientific experiments and data collection. Hence, statement 1 is correct, while statement 2 is incorrect.

122. What is the name of the world's first private flight plan to go on the moon?

- (a) Moon Express (b) Moon Flight
(c) Chandrayaan (d) Moon Mail

Jharkhand P.C.S. (Pre) 2016

Ans. (a)

Moon Express is the world's first private flight plan to go to the moon. Implemented from 2017 this plan was granted assent by Federal Aviation Administration on August 3, 2016.

123. Which country became the first to bring soil from the dark side of the moon in June 2024 ?

- (a) USA (b) China
(c) France (d) India

70th B.P.S.C. (Pre) 2024

Ans. (b)

In June 2024, China's Chang'e 6 mission became the first lunar mission to bring soil from the far side (dark side) of the Moon to Earth. This mission, launched on 3 May, 2024, successfully landed on the Moon on 1 June, 2024 and returned to Earth on 25 June, 2024. Chang'e 6 was the sixth robotic lunar exploration mission by the China National Space Administration (CNSA) and the second CNSA lunar sample-return mission. Like its predecessors in the Chinese Lunar Exploration Program, the spacecraft is named after the Chinese moon goddess Chang'e. It was the first lunar mission to retrieve samples from the far side of the Moon; all previous samples were collected from the near side.

124. Given below are two statements, in which one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : Roscosmos's Luna-25 mission to moon was aimed at ensuring guaranteed access to moon.

Reason (R) : Russia and China together are leading International Lunar Research Station (ILRS) against U.S. led Artemis Accord.

Select the correct answer from the code given below :

Code :

- (a) (A) is true but (R) is false.
(b) Both (A) and (R) are true and (R) is correct explanation of (A).
(c) (A) is false but (R) is true.
(d) Both (A) and (R) are true but (R) is not correct explanation of (A).

U.P. R.O./A.R.O. (Pre) 2023

Ans. (b)

Luna-25 was a failed Russian lunar lander mission by Russian space agency Roscosmos in August, 2023 that planned to land near the lunar south pole. Luna-25 was the first in a series of Russian missions planned as part of the International Lunar Research Station (ILRS), a project initiated by China and Russia to build a base at the lunar south pole by the mid-2030s with the aim of using the rich water-ice resources there for long-term exploration and settlement. Russia and China together are leading ILRS against the U.S. led Artemis Accords. The Artemis Accords is a series of non-binding bilateral arrangements between the United States government and other world governments that elaborates on the norms expected to be followed in outer space. The accords are

related to the Artemis program, an American-led effort to return humans to the Moon by 2026, with the ultimate goal of expanding space exploration to Mars and beyond. Hence, both (A) and (R) are true and (R) is the correct explanation of (A).

125. Which among the following is a recent scientific mission of ISRO for comprehensive study of the Sun?

- (a) Aditya-L1 (b) Bhaskara-L2
(c) Solar Mission-L1 (d) Solar orbiter

M.P. P.C.S. (Pre) 2023

Ans. (a)

On 2 September, 2023, the Polar Satellite Launch Vehicle (PSLV-C57) successfully launched the Aditya-L1 spacecraft, from the Second Launch Pad of Satish Dhawan Space Centre (SDSC), Sriharikota. Aditya-L1 is the first Indian space based observatory to study the Sun from a halo orbit around first Sun-Earth Lagrangian point (L1), which is located roughly 1.5 million km from Earth. Aditya-L1 carries seven scientific payloads indigenously developed by ISRO and national research laboratories including Indian Institute of Astrophysics (IIA), Bengaluru and Inter University Centre for Astronomy and Astrophysics (IUCAA), Pune. On 6 January, 2024, Aditya-L1 was successfully injected on the Halo orbit of Lagrange point1. It will study the solar atmosphere, solar magnetic storms, and their impact on the environment around the Earth.

126. The number of scientific payloads on Aditya-L1 mission for systematic study of Sun is :

- (a) 9 (b) 6
(c) 7 (d) 5

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

Aditya-L1 carries seven payloads on-board to observe the photosphere, chromosphere and the outermost layers of the Sun (the corona) using electromagnetic and particle and magnetic field detectors. Using the special vantage point L1, four payloads (VELC, SUIT, SoLEXS and HELIOS) directly view the Sun (carrying out remote sensing of the Sun) and the remaining three payloads (ASPEX, PAPA and Advanced Tri-axial High Resolution Digital Magnetometers) carrying out in-situ observation studies of particles and fields at the Lagrange point L1, thus providing important scientific studies of the propagatory effect of solar dynamics in the interplanetary medium. The suits of Aditya-L1 payloads are expected to provide most crucial informations to understand the problem of coronal heating, coronal mass ejection, pre-flare and flare activities and their characteristics, dynamics of space weather, propagation of particle and fields etc.

127. In reference to Mangalyaan launched by ISRO, consider the following statements :

- (I) It is also called as the Mars Orbiter Mission.
(II) It was launched in November 2013.

**Select the correct answer using the code given below :
Code :**

- (a) Neither (I) nor (II) (b) Only (II)
(c) Only (I) (d) Both (I) and (II)

U.P. R.O./A.R.O. (Pre) 2023

Ans. (d)

The Mars Orbiter Mission (MOM), or Mangalyaan was a space probe orbiting Mars since 24 September, 2014. It was launched on 5 November, 2013 by the Indian Space Research Organisation (ISRO). It was India's first interplanetary mission and it made ISRO the fourth space agency to achieve Mars' orbit, after Roscosmos, NASA, and the European Space Agency. It made India the first Asian nation to reach the Martian orbit and the first nation in the world to do so on its maiden attempt. It lost communications with Earth in 2022 and then officially declared dead.

128. In which year was India's first, interplanetary mission to planet Mars launched successfully?

- (a) 2013 (b) 2012
(c) 2016 (d) 2015

M.P. P.C.S. (Pre) 2024

Ans. (a)

See the explanation of above question.

129. Consider the following statements :

The Mangalyaan launched by ISRO

1. is also called the Mars Orbiter Mission.
2. made India the second country to have a spacecraft orbit the Mars after the USA.
3. made India the only country to be successful in making its spacecraft orbit the Mars in its very first attempt.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2016

Ans. (c)

Marking India's first venture into the interplanetary space, MOM (Mars Orbiter Mission) or Mangalyaan explored and observed Mars surface features, morphology, mineralogy and the Martian atmosphere. India became the first Asian country (and the first nation in the world to do so in its first attempt) and ISRO became the fourth space agency (after the space agencies of USA, Soviet Union/Russia and European Union) to send a satellite successfully to orbit the Mars. In 2020, UAE and China also sent successful missions to Mars, which were entered orbit of Mars in February, 2021. Hence, option (c) is the correct answer.

130. What is the name of recently launched final module of China's Tiangong Space Station?

- (a) Mengtian
- (b) Tianhe
- (c) Wentian
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Tiangong space station is a permanently crewed space station constructed by China and operated by China Manned Space Agency. The construction of the station is based on the experience gained from its precursors, Tiangong-1 and Tiangong-2. The first module, the Tianhe ("Harmony of the Heavens") core module, was launched on 29 April, 2021. This was followed by multiple crewed and uncrewed missions and the addition of two laboratory cabin modules. The first, Wentian ("Quest for the Heavens"), launched on 24 July, 2022; while the second and final module, Mengtian ("Dreaming of the Heavens") launched on 31 October, 2022.

131. Tianwen-1 is :

- (a) Space Station of China
- (b) Moon rover vehicle of China
- (c) Inter-Continental Missile of China
- (d) Spacecraft of China

R.A.S./ R.T.S. (Pre) 2021

Ans. (d)

Tianwen-1 (TW-1) is an interplanetary mission by the China National Space Administration (CNSA) to send a robotic spacecraft to Mars, consisting of : an orbiter spacecraft, two deployable cameras, lander, remote camera, and the Zhurong rover. The mission was launched from the Wenchang Spacecraft Launch Site on 23 July, 2020 on a Long March 5 heavy-lift launch vehicle. After seven months of transit through the inner Solar System, the spacecraft entered Martian orbit on 10 February, 2021.

132. Which of the following statement is not correct about Chinese space shuttle 'Shenzhou-7'?

- (a) It was manned by four astronauts.
- (b) It was launched by long March 2F rocket.
- (c) For the first time, the Chinese astronauts performed a spacewalk.
- (d) The astronauts returned safely to Earth.

U.P.P.C.S. (Pre) 2008

Ans. (a)

China had launched its third manned spacecraft Shenzhou-7 on September 25, 2008 with three astronauts on board to attempt the country's first-ever space walk. It was blasted off on a Long March II-F Carrier rocket from the Jiuquan satellite launch Centre in the northwestern Gansu province.

Under the China's Shenzhou program, Shenzhou-9 was also launched on June 16, 2012. The mission's crew included the first Chinese female astronaut, Liu Yang.

133. In August 2016, China developed a system nicknamed 'Micius' which is :

- (a) An ancient Chinese musical system
- (b) A quantum satellite communication system
- (c) A computer game
- (d) None of the above

U.P. R.O./A.R.O. (Pre) 2016

Ans. (b)

The world's first quantum communications satellite (Quantum Experiments at Space Scale : QUESS) or Micius was launched by China on 16 August, 2016. The launch of Micius named after the ancient Chinese philosopher and scientist.

134. The 'NISAR satellite' is jointly developed by which of the following organizations?

- (a) ESA and ISRO
- (b) ROSCOSMOS and CNSA
- (c) ISRO and NASA
- (d) ESA and NASA

U.P. P.C.S. (Pre) 2023

Ans. (c)

NISAR (NASA-ISRO Synthetic Aperture Radar) satellite is a joint Earth-observing mission between NASA and the Indian Space Research Organisation (ISRO) with the goal to make global measurements of the causes and consequences of land surface changes using advanced radar imaging. In 2014, NASA and ISRO signed a partnership to collaborate on and launch NISAR. The mission is expected to launch in 2025. NISAR will be the first satellite mission to use two different radar frequencies (L-band and S-band) to measure changes in our planet's surface less than a centimeter across. NASA is providing the mission's L-band synthetic aperture radar, a high-rate communication subsystem for science data, GPS receivers, a solid-state recorder and payload data subsystem. ISRO is providing the spacecraft bus, the S-band radar, the launch vehicle and associated launch services.

135. Recently, ISRO and CNES signed an agreement to set up a joint maritime surveillance system. CNES is a space agency of which country?

- (a) Russia
- (b) Germany
- (c) France
- (d) Canada
- (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

CNES (National Centre for Space Studies; French : Centre National d'études spatiales) is the government space agency of France. ISRO and CNES signed an agreement in March, 2019 to set up a joint maritime surveillance system.

136. Which country launched its first satellite 'Arktika-M' to monitor the climate of the Arctic?

- (a) Japan
- (b) China
- (c) Russia
- (d) India
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (c)

Russia launched its space satellite Arktika-M on 28 February, 2021 on a mission to monitor the climate and environment in the Arctic to expand the country's activities in the region. According to the Russian Federal Space Agency - Roscosmos the satellite can also retransmit distress signals from ships, aircraft or people in remote areas as part of the international Cospas-Sarsat satellite-based search and rescue programme.

137. Which of the following space agencies announced that it had successfully used steam to propel its EQUULEUS spacecraft?

- (a) JAXA
- (b) NASA
- (c) SUPARCO
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

In November 2022, Japan's space agency JAXA has announced that it successfully used steam to propel its water-powered CubeSat EQUilibriUm Lunar-Earth point 6U Spacecraft (EQUULEUS). This was the world's first successful orbit control beyond low-Earth orbit using a water propellant propulsion system. EQUULEUS was one of ten CubeSats launched with the Artemis 1 mission into a heliocentric orbit in cislunar space on the maiden flight of the Space Launch System (SLS) of NASA that took place on 16 November, 2022. EQUULEUS is a nanosatellite of the 6U CubeSat format that will measure the distribution of plasma that surrounds the Earth (plasmasphere) to help scientists understand the radiation environment in that region.

138. Recently which space agency successfully launched the world's smallest satellite-carrying rocket?

- (a) NASA
- (b) ISRO

(c) CNSA

(d) JAXA

R.A.S./R.T.S. (Pre) 2018

Ans. (d)

In February, 2018, 'JAXA' - Japan Aerospace Exploration Agency has successfully launched the smallest-ever rocket to carry a satellite into orbit.

139. NASA's Jet Propulsion Laboratory is located in-

- (a) Houston
- (b) Los Angeles
- (c) Pasadena
- (d) Cape Kennedy

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

NASA's Jet Propulsion Laboratory (JPL) is located in Pasadena, California, United States.

140. What is 'Greased Lightning-10 (GL-10)', recently in the news?

- (a) Electric plane tested by NASA
- (b) Solar-powered two-seater aircraft designed by Japan
- (c) Space observatory launched by China
- (d) Reusable rocket designed by ISRO

I.A.S. (Pre) 2016

Ans. (a)

NASA has developed and successfully flight tested a battery-powered plane with 10 engines that take off and land like a helicopter and fly efficiently like an aircraft. The Greased Lightning or GL-10 prototype successfully transitioned from hover to wing-borne flight during several test flights. It could be used for small package delivery or vertical take off and landing, long endurance surveillance for agriculture, mapping, and other applications.

141. Which spacecraft of NASA has confirmed the presence of water on the Moon?

- (a) ORION
- (b) GEMINI
- (c) SOFIA
- (d) MERCURY

U.P. R.O./A.R.O. (Mains) 2016

Ans. (c)

In a breakthrough discovery, the US space agency NASA confirmed in October 2020, the presence of water on the Moon's sunlit surface for the first time in history. Announcing this discovery about the Moon, the NASA said that its mission, SOFIA (Stratospheric Observatory for Infrared Astronomy), has confirmed the presence of water molecules in Clavius Crater, which is located in the Moon's southern hemisphere. It is notable that India's Chandrayaan-1 was the first Moon mission that indicated the widespread presence of water molecules on lunar soil back in November, 2008.

142. NASA's Deep Impact space mission was employed to take detailed pictures of which comet nucleus ?

- (a) Halley's Comet (b) Hale-Bopp
(c) Hyakutake (d) Temple-1

I.A.S. (Pre) 2007

Ans. (d)

NASA's Deep Impact Mission was launched from Cape Canaveral Air force station on January 12, 2005. It was launched by Delta-II rocket. The Deep Impact spacecraft traveled 429 million Kilometres in 174 days to reach comet Temple-1 and on July 4, 2005 the spacecraft crashed an impactor into Temple-1, allowing scientists to study the icy body's composition. Some previous space missions to comets, such as Giotto and Stardust are just only to examine the surface of comet nuclei. The Deep Impact mission was the first to extract material from a comet's surface.

143. What is the name of NASA vehicle on Mars?

- (a) Phoenix (b) Roger
(c) Florida (d) John Canaday

M.P.P.C.S. (Pre) 2010

Ans. (a)

Phoenix was a robotic spacecraft on a space exploration mission on Mars under the NASA's Mars Scout Program. The Phoenix lander descended on Mars on 25 May, 2008.

144. The first U.S. rover which reached Mars in January 2004 was :

- (a) Discovery (b) Pioneer
(c) Spirit (d) Vanguard

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

Spirit, also known as MER-A (Mars Exploration Rover – A), is a robotic rover on Mars active from 2004 to 2010. It was one of twin rovers of NASA's Mars Exploration Rover Mission. It landed successfully on Mars on January 4, 2004. The second rover, Opportunity (MER-B) landed on the other side of the planet. Robotic rover Curiosity was launched on November 26, 2011 under the NASA's Mars Science Laboratory mission while Perseverance rover was launched on July 30, 2020 as part of NASA's Mars 2020 mission.

145. According to NASA, the evidence for the presence of life on Mars found till today include –

- (a) Water alone (b) Water and gypsum
(c) Water, gypsum and methane (d) None of the above

U.P.P.C.S. (Mains) 2009

Ans. (c)

NASA's Spirit and Opportunity mission on Mars have found the evidence of sulfate which clearly indicated the presence of water in past. Gypsum which is a type of sulfate also found on Mars. But the presence of Methane on Mars has raised the possibility of life on the planet.

146. As per report released by U.S. Journal 'SCIENCE' on 26th July, 2018, a massive underground lake has been detected on which planet?

- (a) Jupiter (b) Saturn
(c) Venus (d) Mars

U.P.P.C.S. (Pre) 2018

Ans. (d)

As per report released by U.S. Journal 'SCIENCE' in July, 2018, astronomers have found a massive underground lake on Mars. It is estimated that this lake is 20 km across and 1.5 km beneath the planet's southern polar ice cap.

147. Methane is present in the atmosphere of :

- (a) Moon (b) Sun
(c) Mars (d) Jupiter

U.P. P.C.S. (Mains) 2016

U.P.P.C.S. (Pre) 2010

Ans. (c) & (d)

The first evidence of methane in the atmosphere of Mars was measured by ESA's Mars Express Orbiter. NASA's Curiosity rover also detected methane presence in the atmosphere of Mars. Jupiter's atmosphere contains mainly Helium and Hydrogen with a trace amount of Water, Ammonia, Methane and other Carbon compounds.

148. The atmosphere of Saturn contains which one of the following gas in large quantity?

- (a) Carbon monoxide (b) Nitrogen
(c) Sulphur dioxide (d) Methane

U.P. P.C.S. (Pre) 2022

Ans. (d)

Saturn is the sixth planet from the Sun and the second-largest planet in our Solar System. Saturn contains approximately 75% hydrogen and 25% helium with traces of other substances like methane, ammonia, water ice etc. Hence, among the given options, option (d) is the most appropriate answer.

149. Which of the following planets has the least density?

- (a) Earth (b) Mars
(c) Venus (d) Saturn
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

Saturn has the least density among all planets of our Solar system.

Planet	Average Density (gm/cm ³)
Earth	5.5
Mercury	5.4
Venus	5.2
Mars	3.9
Neptune	1.6
Jupiter	1.3
Uranus	1.3
Saturn	0.7

150. The fuel tanks of the spacecraft 'Discovery' that was launched into space on July 4, 2006 was filled with –

- (a) Liquid hydrogen only
- (b) Liquid oxygen only
- (c) Mixture of liquid hydrogen and liquid oxygen
- (d) Mixture of liquid hydrogen, liquid oxygen, and liquid helium

U.P.P.C.S. (Pre) 2006

Ans. (c)

The Space Shuttle is a reusable spacecraft that takes off like rocket, orbits the earth like a satellite and then lands like a glider. The space shuttle was made up of three main parts. The orbiter, the external tank and the solid rocket booster. These shuttles are fuelled with liquid hydrogen and liquid oxygen. Challenger, Columbia, Discovery, Atlantis and Endeavour are the space shuttles of NASA, but all of these are retired.

151. Apollo-8 was launched for space exploration on :

- (a) January 6, 1998
- (b) Feb 7, 1998
- (c) March 8, 1998
- (d) April 9, 1998

U.P.P.C.S. (Pre) 1999

Ans. (*)

Apollo-8 was the first mission to take humans to the moon and back. It was launched from Cape Kennedy, Florida, the United States on December 21, 1968.

152. Selene-1, the lunar orbiter mission belongs to which one of the following?

- (a) China
- (b) European Union
- (c) Japan
- (d) USA

I.A.S. (Pre) 2008

Ans. (c)

Selene-1 was Japan's lunar orbiter, which was launched on September 14, 2007. The orbiter's nickname Kaguya was selected by the general public.

153. Consider the following statements in respect of a jet engine and a rocket :

- 1. A jet engine uses the surrounding air for its oxygen supply and so is unsuitable for motion in space.
- 2. A rocket carries its own supply of oxygen in the gas form, and fuel.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither nor 2

I.A.S. (Pre) 2008

U.P.P.C.S. (Pre) 2009

U.P.P.C.S. (Mains) 2007

Ans. (a)

Jet engines and rockets work on the same principle. They produce thrust through an internal pressure difference and as explained by Newton's third law of motion, eject exhaust gases in an equal and opposite direction. The main difference between them is that a jet engine gets the oxygen to burn its fuel from the air while a rocket carries its own supply of oxygen (oxidiser) and fuel in the form of liquid or solid propellants which allows it to operate in space. Hence, statement 1 is correct while statement 2 is incorrect.

154. The jet aircrafts fly very easily and smoothly in the lower stratosphere. What could be the appropriate explanation?

- 1. There are no clouds or water-vapour in the lower stratosphere.
- 2. There are no vertical winds in the lower stratosphere.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2011

U.P.P.C.S. (Mains) 2013

Ans. (c)

Jet aircrafts efficiently manage fuel combustion at low temperature in the stratosphere. The type of turbulence caused by convection heating that causes thunderstorms does not occur in the stratosphere. There are no clouds or water-vapour and also no vertical winds in the lower stratosphere.

155. On which principle, a Jet Engine works?

- (a) Conservation of linear momentum
- (b) Conservation of angular momentum
- (c) Conservation of energy
- (d) Conservation of mass

U.P.R.O./A.R.O. (Mains) 2013

Ans. (a)

Nearly all modern airplanes use jet propulsion to fly. Jet engines and rockets work on the principle of conservation of linear momentum.

156. Rocket works on the principle of the following :

- (a) Bernoulli's theorem
- (b) Energy conservation
- (c) Conservation of Momentum
- (d) Avogadro's concept

U.P.P.C.S. (Mains) 2014

U.P. R.O./A.R.O. (Mains) 2021

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

An electric rocket engine is a jet engine in which material ejected backward (ionized gas in general) causes a forward thrust to the rocket (on the principle of conservation of momentum). The linear momentum (p) or simply momentum of a particle is the product of its mass and its velocity [i.e. $p = mv$].

157. Stratellite is the :

- (a) Uppermost band of stratosphere
- (b) Lowermost band of stratosphere
- (c) High altitude airship positioned in the stratosphere
- (d) Natural satellite of Saturn

U.P. R.O./A.R.O. (Pre) 2016

Ans. (c)

Stratellite is the high altitude airship positioned in the stratosphere and carries one or more repeaters to create wireless communication networks. A stratellite is similar to a satellite, but is stationed in the stratosphere rather than in the orbit.

158. Cryogenic temperature stands for temperatures :

- (a) Below 150°C
- (b) Above 150°C
- (c) Below -150°C
- (d) -273°C

Chhattisgarh P.C.S. (Pre) 2003

Ans. (c)

In Physics, Cryogenics is the study of the production and behaviour of materials at very low temperatures i.e. below -150°C (123 K or -238°F).

159. Low temperatures (cryogenics) find application in :

- (a) Space travel, surgery, and magnetic levitation
- (b) Surgery, magnetic levitation, and telemetry
- (c) Space travel, surgery, and telemetry
- (d) Space travel, magnetic levitation, and telemetry

I.A.S. (Pre) 1999

Ans. (d)

Low temperature (cryogenics) applications are used in space travel, magnetic levitation and telemetry.

160. Cryogenic engines find application in :

- (a) Sub-marine propulsion
- (b) Frost-free refrigerators
- (c) Rocket technology
- (d) Researches in superconductivity

U.P.P.C.S. (Pre) 2006

I.A.S. (Pre) 1995

Ans. (c)

The word Cryogenics was originated from the Greek word 'Kryos' meaning 'frost' and 'genic' to produce. Usually, cryogenics has been used to refer temperatures below approximately -150°C . A cryogenic engine is used as the rocket engine that uses a cryogenic fuel and oxidizer. Its fuel and oxidizer are gases liquefied and stored at very low temperatures (below -150°C).

161. Cryogenic engines use :

- (a) Liquid chlorine as the oxidizer and liquid hydrogen as the fuel.
- (b) Liquid oxygen as the oxidizer and liquid hydrogen as the fuel.
- (c) Liquid fluorine as the oxidizer and liquid hydrogen as the fuel.
- (d) Liquid sulphur dioxide as the oxidizer and liquid hydrogen as the fuel.

U.P.P.C.S. (Mains) 2004

Ans. (b)

The cryogenic engine generally uses liquid oxygen as the oxidizer and liquid hydrogen as the fuel, that require for storage at an extremely low temperature in order to maintain them in a liquid state. Cryogenic engines are also used in space shuttles.

162. The Cryogenic engine is used in :

- (a) Boeing
- (b) Rapid rail engines
- (c) Space shuttle
- (d) Nuclear reactor

U.P.P.C.S. (Pre) 1993

Ans. (c)

See the explanation of above question.

163. Which of the following is not used as rocket propellant –

- (a) Liquid hydrogen
- (b) Liquid oxygen
- (c) Liquid hydrazine
- (d) Kerosene oil

U.P.P.S.C. (GIC) 2010

Ans. (*)

Among the given options, all are used as rocket propellant. Liquid hydrogen, liquid hydrazine and kerosene oil are used as fuel while liquid oxygen is used as oxidiser in rocket propellant. Propellant is a mixture of fuel and oxidiser.

164. The Bharat Electronics Ltd. (BEL) is collaborating with ISRO to develop :

- (a) Agro technology (b) GaAs solar cells
(c) Nuclear technology (d) C-DAC

R.A.S./R.T.S. (Pre) 2018

Ans. (b)

Among the given options, Bharat Electronics Ltd. (BEL) is collaborating with ISRO to develop GaAs (Gallium Arsenide) solar cells. These solar cells for space are more advanced in terms of technology and efficient in order to produce electricity in the space which would eventually help in the longevity of the Indian satellites.

165. Consider the following statements :

1. In the year 2006, India successfully tested a full-fledged cryogenic stage in rocketry.
2. After USA, Russia, and China, India is the only country to have acquired the capability for use of cryogenic stage in rocketry.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (a)

India has become the sixth member of the exclusive club to have developed a cryogenic stage in rocketry with USA, Russia, European space agency, China, and Japan, after successfully testing a full-fledged cryogenic stage for 50 seconds at the Liquid Propulsion Systems Centre (LPSC; now IPRC : ISRO Propulsion Complex) in Mahendragiri, Dist. Tirunelveli, Tamil Nadu in October, 2006.

166. Consider the following statements and find the correct answer from the codes given below :

1. RISAT-2 Satellite was put in orbit by PSLV-C12 in April 2009 from Sriharikota.
2. Another satellite ANUSAT was also put in orbit by the PSLV-C12. This was built by Anna University, Chennai.
3. ANUSAT is an experimental communication satellite.
4. RISAT-2 is surveillance satellite and can keep a watch on country's borders.

Code :

- (a) 1 and 2 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1, 2, 3, and 4 all

U.P.P.C.S. (Mains) 2008

Ans. (d)

RISAT-2 or Radar Imaging Satellite-2 is an Indian radar reconnaissance satellite. It was successfully launched by PSLV-C12 rocket on 20 April, 2009 from Satish Dhawan Space Centre, Sriharikota. It was designed to monitor India's border. ANUSAT is an experimental communication satellite designed and developed by Anna University, Chennai. It was successfully launched aboard PSLV-C12 along with RISAT-2 from the same place.

167. With reference to Indian satellites and their launchers, consider the following statements :

1. All the INSAT series of satellites were launched abroad.
2. PSLVs were used to launch IRS-series of satellites.
3. India used the indigenously built cryogenic engines for the first time for powering the third stage of GSLV.
4. GSAT, launched in the year 2001, had payloads to demonstrate digital broadcast and internet services.

Which of these statements are correct?

- (a) 1, 2, 3 and 4 (b) 2, 3 and 4
(c) 1, 2 and 4 (d) 1 and 3

I.A.S. (Pre) 2002

Ans. (b)

GSLV is a three-stage launcher, whose third stage used an indigenous- built cryogenic engines. This indigenous engine was built by ISRO's Liquid Propulsion System Centre (Present name : ISRO Propulsion Complex) at Mahendragiri in Tamil Nadu. GSAT-1 satellite was launched on 18 April, 2001 which had payloads to demonstrate digital broadcasts and internet services. PSLVs were used to launch IRS series of satellites and apart from this, it is also used to launch various other satellites. INSAT-4C was launched from Sriharikota on 10 July, 2006. So in the present context, the statement 2, 3, and 4 are correct but 1 is false.

168. With reference to India's satellite launch vehicles, consider the following statements :

1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-stage launch vehicle with the first and third stages using solid rocket motors; and the second and fourth stages using liquid rocket engines.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3
(c) 1 and 2 (d) 3 only

I.A.S. (Pre) 2018

Ans. (a)

PSLV (Polar Satellite Launch Vehicle) is an indigenously developed expendable launch system of the ISRO. It comes in the category of medium-lift launchers with a reach up to various orbits, including the Geo Synchronous Transfer orbit, Lower Earth orbit and Polar Sun Synchronous orbit. GSLV is also operated by ISRO. It is used to launch a satellite into Geo Synchronous Transfer orbit. The GSLV is designed mainly to deliver the communication satellites to the height of 36000 km. PSLV launches mainly the earth monitoring satellites to the Polar orbits and these satellites do not appear to fix in the same position in the sky. GSLV Mk III (LVM3) is a three-stage launch vehicle with the first stage using a solid propellant, second stage using liquid propellant and the third stage is the cryogenic stage. Hence, only statement 1 is correct.

169. An artificial satellite can be tracked very precisely from the Earth by using :

- (a) Doppler effect (b) Radar
(c) Sonar (d) Pulsar

U.P.P.C.S. (Pre) 2017

Ans. (a)

An artificial satellite can be tracked very precisely from the Earth by using Doppler effect. It is done with the help of tracking transmitter. With the help of doppler effect, this tracking is possible.

170. Which one of the following is not a space satellite?

- (a) SLV-3 (b) RS-D1
(c) IRS-1D (d) INSAT-2D

U.P.P.S.C. (GIC) 2010

U.P. Lower Sub. (Spl.) (Pre) 2004

Ans. (a)

Rohini is a series of satellites launched by the Indian Space Research Organisation. The Rohini series consist of four satellites, all of which were launched successfully by ISRO. RS-D1 was launched on 31 May, 1981. IRS Satellites is a series of Indian Remote Sensing Satellite for earth's observation. IRS-ID was launched on 29 September, 1997. INSAT-2D satellite was launched on 4 June, 1997. SLV-3 is not a space satellite, it is a satellite launch vehicle which was developed by ISRO.

171. GSLV-D2 which has been recently launched into the orbit is-

- (a) Defence Satellite
(b) Exploration Satellite
(c) Communication Satellite
(d) Traffic Satellite

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (*)

GSLV is India's indigenous satellite launch vehicle. It was primarily developed to launch INSAT class of satellites into geosynchronous transfer orbit. GSLV-D2 was the second developmental test flight (8 May, 2003) of this vehicle. In this flight GSLV had placed a heavier satellite, the 1800 kg GSAT-2 into geosynchronous transfer orbit.

172. Consider the following statements:-

- Vigyan Rail is a Science Exhibition on wheels organized by the Council of Scientific and Industrial Research.**
- Vigyan Prasar is an autonomous body under the ministry of human resources development.**
- The educational satellite of ISRO (EDUSAT) was launched in 2004 from French Guiana.**

Which of the above statement is correct

- (a) 1, 2, and 3 (b) 1 and 2
(c) 2 and 3 (d) None of the above

I.A.S. (Pre) 2005

Ans. (d)

Vigyan Rail is a science exhibition on the wheels organized jointly by Ministry of Railway and the Vigyan Prasar under the Department of Science and Technology. It was started on 15 December, 2003 for depicting India's achievements in various fields of science and technology. Therefore Statement 1 is false. Vigyan Prasar is an autonomous organization under the Department of Science and Technology and not under the Ministry of Human Resources and Development. It was established in the year 1989. So statement 2 is also false. EDUSAT is the satellite exclusively devoted to meet the demands of the education sector. It was launched on 20 September, 2004 from Satish Dhawan Space Centre, Sriharikota, Andhra Pradesh and not from French Guiana. Eventually, the Statement 3 is also false.

173. India became the first country to send Educational Services Satellite in space, Edusat. When was the satellite launched ?

- (a) 20th September, 2003 (b) 20th September, 2004
(c) 4th July, 2004 (d) 20th December, 2003

U.P.P.C.S. (Mains) 2004

Ans. (b)

See the explanation of above question.

174. Consider the following statements :

- India launches its first full-fledged meteorological satellite (METSAT) in September, 2002.**
- For the first time, the space vehicle PSLV C-4 carried a payload of more than 1000 kg into a geosynchronous orbit.**

Which of these statements is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2003

Ans. (c)

India's first full-fledged meteorological satellite 'METSAT' was launched on 12 September, 2002. This was the first time the space vehicle PSLV-C4 has entered into the geosynchronous orbit with having more than 1000 Kg (1060 kg) lift-off mass. METSAT was later renamed Kalpana-I to honour the late Indian-born astronaut Kalpana Chawla.

175. Kalpana-I is the new name given to one of the following :

- (a) INSAT-1A (b) METSAT
(c) INSAT-2B (d) Agni-III

U.P.P.C.S. (Mains) 2004

Ans. (b)

See the explanation of above question.

176. In September 2002, India launched :

- (a) GSAT (b) INSAT-3B
(c) IRS-P4 (d) METSAT

U.P.P.C.S. (Pre) 2003

Ans. (d)

See the explanation of above question.

177. Which of the following is Academic Institute Satellite?

- (a) CARTOSAT-2B (b) KALPANA-1
(c) INSAT-2E (d) SATYABAMASAT

M.P.P.C.S. (Pre) 2019

Ans. (d)

Sathyabamasat is a micro experimental satellite developed by students and faculty of Sathyabama University, Chennai to collect data on greenhouse gases. It was launched along with the Cartosat-2C satellite atop PSLV-C34 on 22 June, 2016.

178. India's advanced communication satellite GSAT-8 was launched on May 21, 2011 from :

- (a) Hassan (b) Kourou
(c) Shriharikota (d) Thiruvananthapuram

U.P.P.C.S. (Mains) 2010

Ans. (b)

India's advanced communication satellite GSAT-8 was successfully launched on 21 May, 2011 by the Ariane-V launch vehicle of Arianespace from Kourou, French Guiana. Ariane-V placed GSAT-8 into the intended geosynchronous transfer orbit of 35,861 km apogee and 258 km perigee, with an orbital inclination of 2.503 deg with respect to the equator.

179. The Indian GSAT-7 satellite is exclusively meant for-

- (a) Weather forecast (b) Military communication

(c) Mars mission

(d) Special coverage of moon

R.A.S./R.T.S.(Pre) 2013

Ans. (b)

GSAT-7 or INSAT-4F is a multi-band communication satellite developed by ISRO. It is also known as 'RUKMINI'. The multiband home-built satellite is totally dedicated to Indian Navy. It was successfully launched on 30 August, 2013 by Ariane-5 launch vehicle of Arianespace from Kourou French Guiana.

180. List of few satellites manufactured by ISRO is given below along with the date of their launch. Which two of the following Satellites failed to reach the orbit?

Satellite	Launch Date
1. GSAT-2	– May 8 th , 2003
2. GSAT-4	– 15 th April, 2010
3. GSAT-5P	– 25 th December, 2010
4. GSAT-12	– 15 th July, 2011

Code :

- (a) 4 and 1 (d) 1 and 2
(c) 2 and 3 (d) 3 and 4

R.A.S./R.T.S. (Pre) 2013

Ans. (c)

GSAT-4 was an experimental communication satellite launched by ISRO in April, 2010 which failed to reach to orbit after the rocket's third stage malfunctioned. GSAT-5P was the fifth satellite launched in the GSAT series. It was an exclusive communication satellite to further argument the communication services currently provided by the Indian National Satellite system. Weighing 2310 at the lift, GSAT-5P carried 24 Normal C-band and 12 extended C-band transponders. However, GSAT-5P was not placed in orbit as GSLV-F06 could not complete its mission.

181. Which one of the following satellites was successfully launched by India on 29th September, 2012 ?

- (a) GSAT-6 (b) GSAT-9
(c) GSAT-10 (d) GSAT-12

U.P.P.C.S. (Mains) 2012

Ans. (c)

On 29th September, 2012 India successfully launched the heaviest telecommunication satellite GSAT-10 by the Ariane-5 rocket from Guiana Space Centre.

182. ISRO has successfully launched GSAT-6A satellite from which rocket vehicle?

- (a) GSLV-F07 (b) GSLV-F09